

Reviewed Preprint

v1 • July 17, 2025

Not revised

Reviewed Preprint

v2 • November 28, 2025

Revised by authors

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Competing interests: JWK and RLW are shareholders and scientific advisory board members of Protego Biopharma, which has licensed ATF6 activators for therapeutic development. TM is a scientific advisory board member of Cure GABA- A Variants Foundation, a nonprofit organization.

Funding: See [page 25](#)

Reviewing editor: Johannes Herrmann, University of Kaiserslautern, Germany

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Phenylhydrazone-based Endoplasmic Reticulum Proteostasis Regulator Compounds with Enhanced Biological Activity

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eLife Assessment

This study reports the **important** development and characterization of next-generation analogs of the molecule AA263, which was previously identified for its ability to promote adaptive ER proteostasis remodeling. The evidence supporting the conclusions is **convincing**, with rigorous assays used to benchmark the changes in potency and efficacy of the AA263 analogs as well as AA263 targets. The ability of AA263 analogs to restore the loss of function associated with disease-associated proteins prone to misfolding will be of interest to pharmacologists, chemical biologists, and cell biologists, as well as those working on protein misfolding disorders.

<https://doi.org/10.7554/eLife.107000.2.sa4>

Abstract

Pharmacological enhancement of endoplasmic reticulum (ER) proteostasis is an attractive strategy to mitigate pathology linked to etiologically-diverse protein misfolding diseases. However, despite this promise, few compounds have been identified that enhance ER proteostasis through defined mechanisms of action. We previously identified the phenylhydrazone-based compound AA263 as a compound that promotes adaptive ER proteostasis remodeling through mechanisms including preferential activation of the ATF6 signaling arm of the unfolded protein response (UPR). However, the protein target(s) of AA263 and the potential for further development of this class of ER proteostasis regulators had not been previously explored. Here, we employ chemical proteomics to demonstrate that AA263 covalently targets a subset of ER protein disulfide isomerases, revealing a potential molecular mechanism for the activation of ATF6 afforded by this compound. We then use medicinal chemistry to establish next-generation AA263 analogs showing improved potency and efficacy for ATF6 activation, as compared to the parent compound. Finally, we show that treatment with these AA263 analogs enhances secretory pathway proteostasis to correct the pathologic protein misfolding and trafficking of both a destabilized, disease-associated α 1-antitrypsin (A1AT) variant and an epilepsy-associated GABA_A receptor variant. These results establish AA263 analogs with enhanced potential for correcting imbalanced ER proteostasis associated with etiologically-diverse protein misfolding disorders.

Introduction

Nearly one-third of the human proteome is targeted to the endoplasmic reticulum (ER) for folding and trafficking to downstream secretory environments including the plasma membrane, lysosome, and the extracellular space.^{1–3} The folding and trafficking versus degradation of these

secretory proteins is decided by a process termed ER quality control, wherein ER-localized chaperones and folding factors engage proteins within the ER to facilitate their folding into a folded, trafficking-competent conformation.^{1,2,4} Proteins unable to attain a folded conformation in the ER through interactions with ER chaperones are instead recognized by ER-localized degradation factors that promote their targeting to degradation pathways such as ER-associated degradation by the proteasome or ER-phagy accomplished by the lysosome.⁴ Through this ER quality control process, cells promote the trafficking of folded, functional proteins through the secretory pathway and prevent the accumulation of non-native or aggregation-prone conformations within the ER or in downstream secretory environments.^{5,6}

Despite the efficacy of ER quality control, failure of this process is implicated in the onset and pathogenesis of numerous, etiologically-diverse diseases, collectively referred to as protein misfolding diseases.⁷ These diseases are primarily associated with destabilizing mutations within a secretory protein that hinder ER quality control machinery, ultimately leading to toxic protein aggregation and/or loss of function pathologies. For example, the aberrant secretion and toxic extracellular aggregation of destabilized, aggregation-prone variants of amyloidogenic proteins including transthyretin (TTR) or immunoglobulin light chain (LC) are implicated in the onset and pathogenesis of protein aggregation diseases such as the TTR amyloidoses and Light Chain Amyloidosis.^{8,9} Alternatively, mutations in subunits of the gamma-aminobutyric acid type A (GABA_A) receptors prevent the proper folding, assembly and trafficking of these proteins to the plasma membrane. This loss-of-function phenotype has been implicated in numerous neurological disorders, including genetic epilepsy and neurodevelopmental delay.^{10,11} Other proteinopathies, such as alpha-1-antitrypsin (AAT) deficiency can involve both toxic protein aggregation and loss of function pathologies. In this disease, destabilized AAT variants (e.g., PIZZ) can lead to both gain of toxic function pathology through intra-ER retention causing liver pathology and loss of function pathology associated with poorly functional secreted AAT leading to chronic obstructive pulmonary disease (COPD) in the lung.^{12,13}

The central involvement of ER quality control in the onset and pathogenesis of protein misfolding diseases has suggested that developing strategies to enhance ER quality control offers a potential opportunity to broadly mitigate pathologies linked to the aberrant folding or degradation of proteins targeted to the ER.¹⁴ One attractive strategy to regulate ER quality control is through targeting the unfolded protein response (UPR) – the predominant stress responsive signaling pathway that regulates ER quality control in response to pathologic ER insults.^{15–17} The UPR comprises three integrated signaling pathways activated downstream of the ER membrane proteins IRE1, PERK and ATF6.¹⁸ ER stress activates the UPR leading to downstream transcriptional and translational signaling that functions to both alleviate the ER stress and enhance ER function. ER quality control is primarily regulated by the UPR-associated transcription factors XBP1s (downstream of IRE1) and ATF6 (a cleaved product of full length ATF6).^{19,20} These transcription factors induce expression of overlapping, but distinct, sets of ER chaperones, folding enzymes, and degradation factors that remodel the composition and the capacity of ER quality control pathways by creating emergent functions.^{21,22} Consistent with this, stress-independent activation of XBP1s, or especially, ATF6 has been shown to correct pathologic imbalances in ER quality control implicated in the pathogenesis of many different protein misfolding diseases, including those indicated above.^{23–25}

The potential to improve ER quality control in protein misfolding diseases through activation of ATF6 led to a significant interest in developing pharmacologic approaches that selectively induce activation of this UPR transcriptional program. Towards that aim, we previously used high throughput screening to identify first-in-class ER proteostasis regulator compounds, such as compound AA147, that preferentially activate the ATF6 signaling arm of the UPR.²⁶ We showed that AA147 activates ATF6 through a mechanism involving metabolic conversion of the prodrug AA147 to an electrophilic quinone methide on the ER membrane that covalently modifies a subset of ER protein disulfide isomerases (PDIs).²⁷ This leads to an increase in reduced ATF6 that can traffic to the Golgi for proteolytic release of the active ATF6 transcription factor domain.^{28–31} Intriguingly, treatment with AA147 improves ER quality control for many different destabilized,

disease-associated proteins linked to protein aggregation and loss-of-function protein misfolding diseases including AAT deficiency and GABA_A receptor-associated epilepsy, as well as protein aggregation-associated degenerative diseases such as TTR amyloidosis and immunoglobulin light chain amyloidosis.^{26,32–34} As predicted based on the mechanism of action of AA147, these benefits can be attributed to ER proteostasis remodeling induced by either direct PDI modification and/or by ATF6 activation, demonstrating the broad potential for this approach to influence ER proteostasis of multiple different disease-relevant proteins.^{35,36} However, despite this promise, the continued development of AA147 has proven challenging, as AA147 has been shown to be largely recalcitrant towards scaffold modifications aimed at improving its potency, selectivity, and efficacy for ATF6 activation.^{27,37}

In addition to finding AA147, our original high-throughput screen also identified the phenylhydrazone compound AA263 as a compound that preferentially activates the ATF6 arm of the UPR.²⁶ However, the mechanism of AA263-dependent ATF6 activation and the potential for the development of improved analogs based on this scaffold by medicinal chemistry have not been previously explored. Herein, we show that AA263, like AA147, covalently targets ER PDIs, providing a potential mechanism to explain the preferential ATF6 activation afforded by this compound. Intriguingly, we also show that modification of the AA263 B-ring at the *para* position allows for the establishment of compounds with improved selectivity and efficacy for ATF6 activation. Taking advantage of this, we establish next generation AA263 analogs that show improved potency and efficacy in comparison to the parent AA263. Further, we demonstrate that these enhanced AA263 analogs correct ER quality control defects in cellular models expressing disease-relevant proteins including the Z-variant of AAT and epilepsy-associated variants of GABA_A receptors. Collectively, these results establish AA263 and its associated analogs as ER proteostasis regulators with enhanced potential for further development, expanding the toolbox of compounds suitable for targeting ER quality control in the context of etiologically diverse protein misfolding diseases.

Results

AA263 covalently modifies a subset of ER PDIs

ER proteostasis regulators including AA147 and AA132 activate the ATF6 arm of the UPR through a mechanism involving their metabolic activation to a quinone methide and subsequent covalent targeting of ER PDIs.^{27,38} Intriguingly, 2-hydroxyphenylhydrazones are known to tautomerize to a quinone methide (AA263-QM), suggesting that AA263 could similarly activate ATF6 through spontaneous quinone methide electrophile generation and covalent PDI targeting (**Fig. 1A**), without the requirement of metabolic activation.³⁹ Interestingly, co-treatment with β -mercaptoethanol (BME), which acts as both an exogenous nucleophile and modulator of cellular redox potential, or the antioxidant resveratrol decreased AA263-dependent activation of the ERSE-FLuc reporter (**Fig. 1B,C**) suggesting that an ER enzyme oxidative activation of AA263 to an unknown electrophile could also be playing a role in PDI conjugation. Consistent with this, synthesis of an AA263 analog lacking the hydroxyl moiety on the A-ring (AA263-1; **Fig. S1A**) does not activate the ATF6-selective ERSE-FLuc reporter in HEK293 cells. These results support a model whereby AA263 activates ATF6 through a mechanism involving both enzymatic oxidation and direct tautomerization to an AA263-QM and subsequent covalent modification of predominantly ER protein targets (**Fig. 1A**).

To monitor covalent protein labeling by AA263, we generated an AA263 analog with replacement of the nitro group on the B-ring with an alkyne amenable for affinity enrichment experiments (AA263^{yne} **Fig. 1D**). We found that this analog robustly activated the ERSE-FLuc reporter, demonstrating an ~2-fold increase in potency, as compared to AA263 (**Fig. 1E**). Co-treatment with the selective ATF6 inhibitor Ceapin-A7 (CP7)^{40–42} blocked AA263^{yne}-dependent increases in ERSE-FLuc activity, confirming that this effect can be attributed to ATF6 activation (**Fig. S1B**). We also confirmed that the AA263^{yne}-dependent activation of the ERSE-FLuc reporter was inhibited by co-treatment with BME or resveratrol (**Fig. 1F**). We treated HEK293 cells with AA263^{yne} and

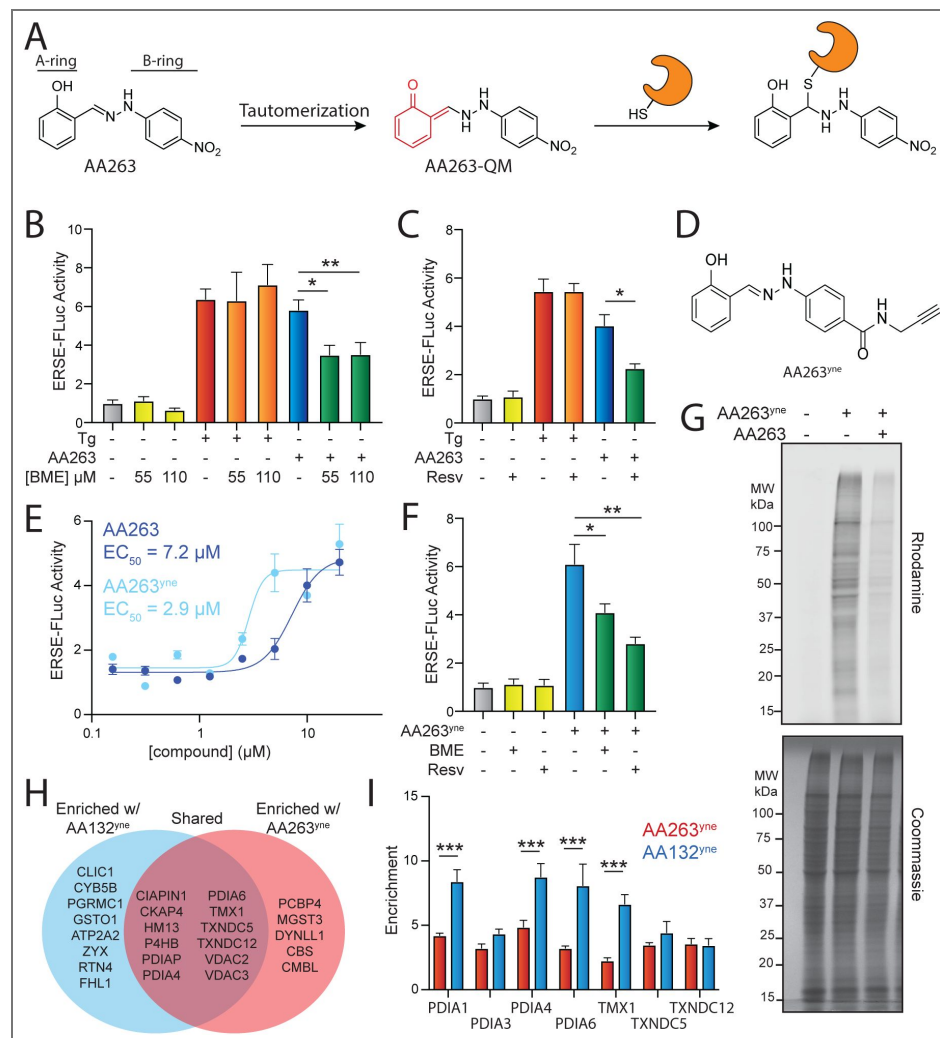


Figure 1. AA263 covalently modifies protein disulfide isomerase family members.

A. Mechanism of AA263 metabolic activation and covalent protein modification **B.** Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated for 18 hr with AA263 (10 μ M) or thapsigargin (Tg, 500 nM) in the presence or absence of β -mercaptoethanol (BME; 55 μ M or 110 μ M). Error bars show SEM for N>6 replicates. * p <0.05, ** p <0.01 for unpaired t-test. **C.** Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated for 18 hr with AA263 (10 μ M) or Tg (500 nM) in the presence or absence of resveratrol (2.5 μ M). Error bars show SEM for N>6 replicates. * p <0.05 for unpaired t-test. **E. D.** Structure of AA263yne **E.** Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated for 18 h with the indicated dose of AA263 or AA263yne. Error bars show SEM for n =3 replicates. The EC_{50} is shown. **F.** Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated with AA263yne (10 μ M) in the presence or absence of BME (55 μ M) or resveratrol (2.5 μ M). * p <0.05, ** p <0.01 for unpaired t-test. **G.** Representative SDS-PAGE gel of Cy5-conjugated proteins from HEK293T cells treated for 4 h with vehicle (0.1% DMSO), AA263yne (5 μ M), or the combination of AA263yne (5 μ M) and AA263 (20 μ M). Coomassie stained gel is shown below. **H.** Venn diagram of identified targets of AA132yne and AA263yne. Hits defined as proteins with a significant fold change greater than 3 (p <0.01) that were identified in two independent biological experiments. **I.** TMT reporter ion enrichment ratio of select PDIs from comparative chemoproteomic experiment in HEK293T cells treated with the indicated compound relative to DMSO (n = 8 biological replicates). *** p <0.005 for a two-way ANOVA.

visualized protein modification by appending a Cy5-azide fluorophore to the terminal alkyne of conjugated proteins using Cu(I)-catalyzed azide-alkyne cycloaddition (CuAAC). AA263^{yne} showed robust protein labeling in HEK293 cells upon in site compound treatment (**Fig. 1G**). Co-treatment with excess AA263 reduced labeling by AA263^{yne}, confirming that AA263^{yne} and AA263 label the same subset of the human proteome. Further, co-treatment with BME or resveratrol also reduced AA263^{yne} proteome labeling (**Fig. S1C**). Intriguingly, unlike AA147^{yne}, treatment of lysates with AA263^{yne} showed mild proteome labeling, indicating that some AA263-dependent protein labeling may proceed through direct tautomerization to the electrophilic QM species (**Fig. S1D**).

We next sought to identify proteins modified by AA263 using Tandem Mass Tag (TMT)-based quantitative proteomics. We previously showed that preferential ATF6 activation afforded by compounds such as AA147 and AA132 is attributed to the extent of compound-dependent modification of PDIs.³⁸ Thus, we compared the extent of proteome labeling observed with AA263^{yne} labeling to that observed with AA132^{yne} – a compound that globally activates the UPR through covalently modifying a large number of PDIs.³⁸ For these comparative chemoproteomic experiments, we treated HEK293 cells with vehicle, AA263^{yne} or AA132^{yne} and then used CuAAC to append biotin-azide to modified proteins. Proteins were then enriched with streptavidin, digested with trypsin digestion, labeled with tandem mass tag (TMTs), and analyzed by LC-MS/MS.^{43,44} Modified proteins were defined by a 3-fold enrichment in compound-treated cells ($p < 0.01$), as compared to vehicle-treated cells. We identified 17 proteins covalently modified by AA263^{yne}, as compared to 20 proteins modified by AA132^{yne} (**Fig. 1H**). Intriguingly, 12 proteins were shared between these two conditions, including 7 different ER-localized PDIs (**Fig. 1H**). This includes PDIs previously shown to regulate ATF6 activation including TXNDC12/ERP18.^{45,46} These results are similar to those observed when comparing proteins modified by the preferential ATF6 activating compound AA147^{yne} and AA132^{yne}.³⁸ Further, we found that the extent of labeling for PDIs including PDIA1, PDIA4, PDIA6, and TMX1, but not TXNDC12, showed greater modification by AA132^{yne}, as compared to AA263^{yne} (**Fig. 1I**). Similar results were observed for AA147^{yne}.³⁸ This suggests that, like AA147, the preferential activation of ATF6 afforded by AA263 is likely attributed to the modifications of a subset of multiple different ER-localized PDIs by this compound. Four of the 5 proteins that formed conjugates with AA263^{yne}, but not AA132^{yne}, are cytosolic proteins, suggesting that tautomerization to an AA263-QM may be occurring as a minor quinone methide formation pathway. However, spontaneous tautomerization to the quinone methide of AA263 seems unlikely to be the main pathway of electrophile formation from AA263 given the remarkable preference of AA263 to react with ER-localized proteins, which is not a general feature of other small molecule electrophiles.^{47,48} This specificity for ER proteins instead suggests the localized generation of AA263 quinone methides at the ER membrane, likely through metabolic activation by different ER localized oxidases, which has previously been shown to contribute to the selective modification of ER proteins afforded by other compounds such as AA147.⁴⁹

Structure activity relationship studies identify AA263 analogs with improved potency and efficacy

The increased potency observed for AA263^{yne}, as compared to AA263, suggests that the B-ring of AA263 may be amenable to medicinal chemical modification for modulating the activation of ATF6 by this class of compounds (**Fig. 1E**). Further, nitro groups are known to alter mitochondrial function, so removal of this functional group could improve cellular tolerability and reduce cytotoxicity associated with two successive 1 electron reductions of the nitro group to afford cytotoxic aromatic nitroso compounds.^{50,51} Towards that aim, we synthesized a panel of AA263 derivatives comprising a heteroaromatic B ring and aryl B rings with varying functional groups largely but not exclusively appended at the para-position of the B ring (**Fig. 2A**). We screened these compounds for ATF6 activation using HEK293 cells stably expressing the ATF6-selective ERSE-Fluc reporter (**Fig. 2B**). This effort demonstrated that ATF6 activators are tolerant of B-ring *para* amide and ester functional groups, reflected by AA263^{yne} and AA263-5 being the two

most active AA263 analogs. Interestingly, the *para* carboxylate substitution affords an inactive compound (AA263-4). We confirmed that AA263^{yne} and AA263-5 induced expression of the ATF6 target gene *HSPA5/BiP* in HEK293 cells (**Fig. S2A**) to levels higher than that observed for AA263, without increasing expression of the IRE1/XBP1s target gene *ERdj4* (**Fig. S2A**) or the PERK/ISR target gene *CHOP/DDIT3* (**Fig. S2A**). The inactive compound AA263-2 (**Fig. 2A**) featuring a heterocyclic B-ring was used as a control, and did not show activation of UPR target genes. ATF6 activators AA263^{yne} and AA263-5 also both showed increased potency for ERSE-FLuc activation relative to AA263, further highlighting the improved activity of these two compounds (**Fig. S2B**).

Next, we used RNAseq to further probe the selectivity for ATF6 activation in HEK293 cells treated with either vehicle, AA263, AA263^{yne} or AA263-5. Initially, we monitored the activation of ATF6 and other arms of the UPR by comparing the expression of sets of genes regulated by these different UPR signaling pathways.⁵² Aligning with our qPCR results, AA263^{yne} and AA263-5 increased expression of ATF6 target genes, in comparison to AA263 (**Fig. 2C**, Table S1). These compounds also showed a modest increase in IRE1/XBP1s target genes, reflecting the known overlap between ATF6 and IRE1/XBP1 target genesets.²² In contrast, both compounds modestly reduced expression of PERK/ISR target genes. This may result from removal of the nitro group, as this geneset can be induced by mitochondrial dysfunction resulting from the presence of this moiety.^{53,54} These results indicate that AA263^{yne} and AA263-5 both increase adaptive ATF6 signaling, while maintaining or improving preferential selectivity for this specific arm of the UPR over the other two UPR arms. Importantly, we did not observe activation of genesets regulated downstream of other stress-responsive signaling pathways such as the oxidative stress response (OSR) or heat shock response (HSR) in HEK293 cells treated with these other AA263 analogs (**Fig. S2C**, Table S1). Further, Gene Ontology (GO) analysis showed that AA263^{yne} and AA263-5 primarily induced expression of genes involved in biological pathways linked to ER proteostasis and UPR activation (Table S2).⁵⁵ Collectively, these results indicate that AA263^{yne} and AA263-5 both show enhanced activation of the ATF6 transcriptional program, without sacrificing the transcriptome wide preferential selectivity for ATF6 activity observed for the parent compound AA263.

We next sought to evaluate whether we could further elaborate the *para*-amide substructure of the AA263^{yne} scaffold by a medicinal chemistry strategy to provide additional enhancement of ATF6 activity. Towards that aim, we synthesized AA263^{yne} analogs where the alkyne moiety was replaced with various aliphatic or aromatic substituent groups (**Fig. 3A**). We then monitored activation of the ERSE-FLUC ATF6 reporter stably expressed in HEK293 cells treated with increasing doses of these analogs. This identified two analogs, AA263-15 and AA263-20, that showed higher levels of reporter activation, as compared to AA263 or AA263^{yne}, with exhibiting increased potency (**Fig. 3B,C**). Both AA263-15 and AA263-20 increased expression of the ATF6 target gene *BiP* at 6 h after 5 μ M treatment to levels greater than that observed for AA263^{yne} in HEK293 cells (**Fig. 3D**). However, these compounds did not significantly influence expression of the IRE1/XBP1s target gene *ERdj4* or the PERK target gene *CHOP* in these cells (**Fig. S3A**). Further, we confirmed that both AA263-15 and AA263-20 competed with proteome labeling afforded by AA263^{yne} (**Fig. S3B**), demonstrating that these compounds covalently targeted a similar subset of the proteome. These results demonstrate the potential for further developing AA263 analogs for preferential ATF6 activation and identify two compounds, AA263-15 and AA263-20 as compounds with improved activity relative to AA263 or AA263^{yne}.

Enhanced AA263 analogs improve secretory proteostasis for the disease-associated Z-variant of A1AT

We next sought to define the potential of AA263 analogs for promoting adaptive ER remodeling in cellular models of protein misfolding diseases. AAT deficiency is a complex, multi-tissue disorder caused by the pathologic aggregation of destabilized AAT variants in the ER of hepatocytes comprising the liver and the reduced ability of secreted mutant AAT to inhibit neutrophil elastase, leading to lung dysfunction.¹² Previously, we showed that treatment with ER proteostasis factors such as AA263 reduced intra- and extracellular AAT polymers and improved neutrophil elastase

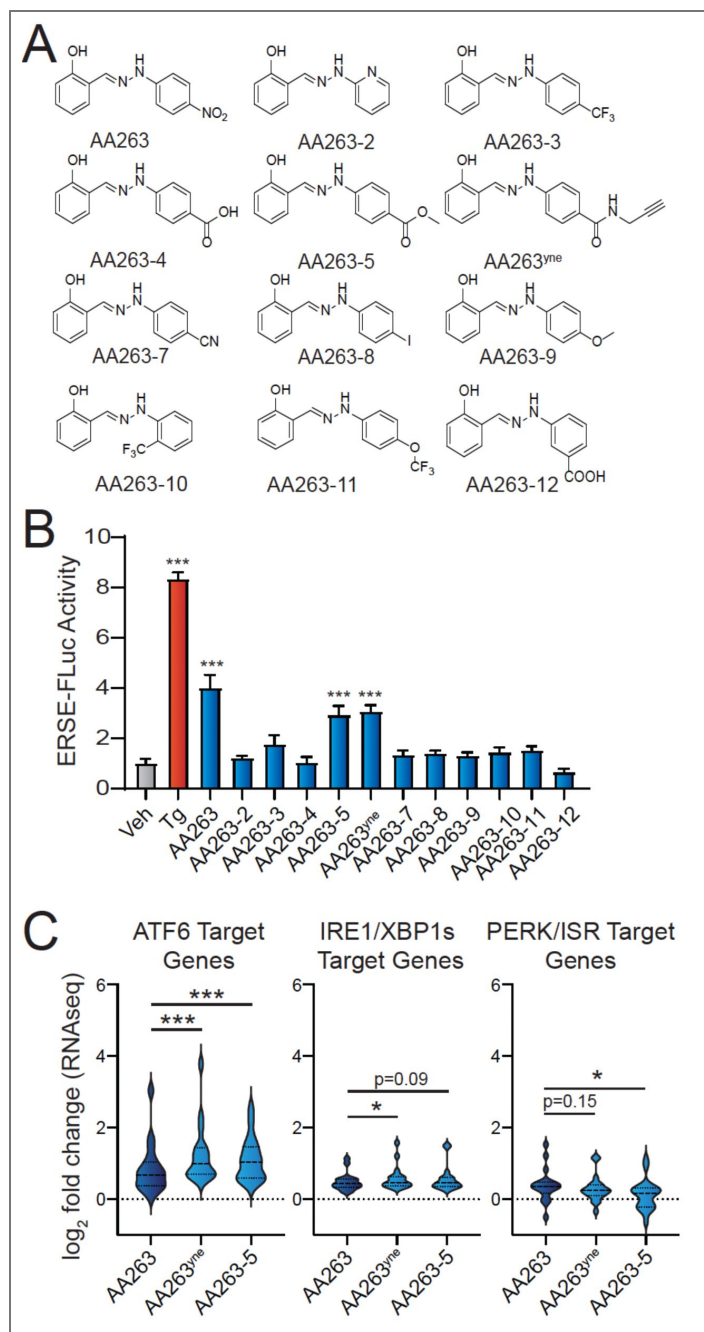


Figure 2. Identification of AA263 analogs that show enhanced ATF6 activation.

A. Structures of AA263 analogs. **B.** Activation of the ERSE.Fluc ATF6 reporter in HEK293T cells reporter treated for 18 h with vehicle, Tg (0.5 μ M), or the indicated analog (10 μ M). Error bars show SEM for n=3-6 biological replicates. ***p<0.005 from one-way ANOVA. **C.** Expression, measured by RNAseq, of gene sets comprising target genes regulated downstream of the ATF6 (left), IRE1/XBP1s (middle), or PERK/ISR (right) arms of the UPR in HEK293T cells treated for 6 h with 10 μ M AA263, AA263^{yne}, or AA263-5. Full RNAseq data and genesets used in this analysis are shown in Table S1. *p<0.05, ***p<0.005 for one-way ANOVA.

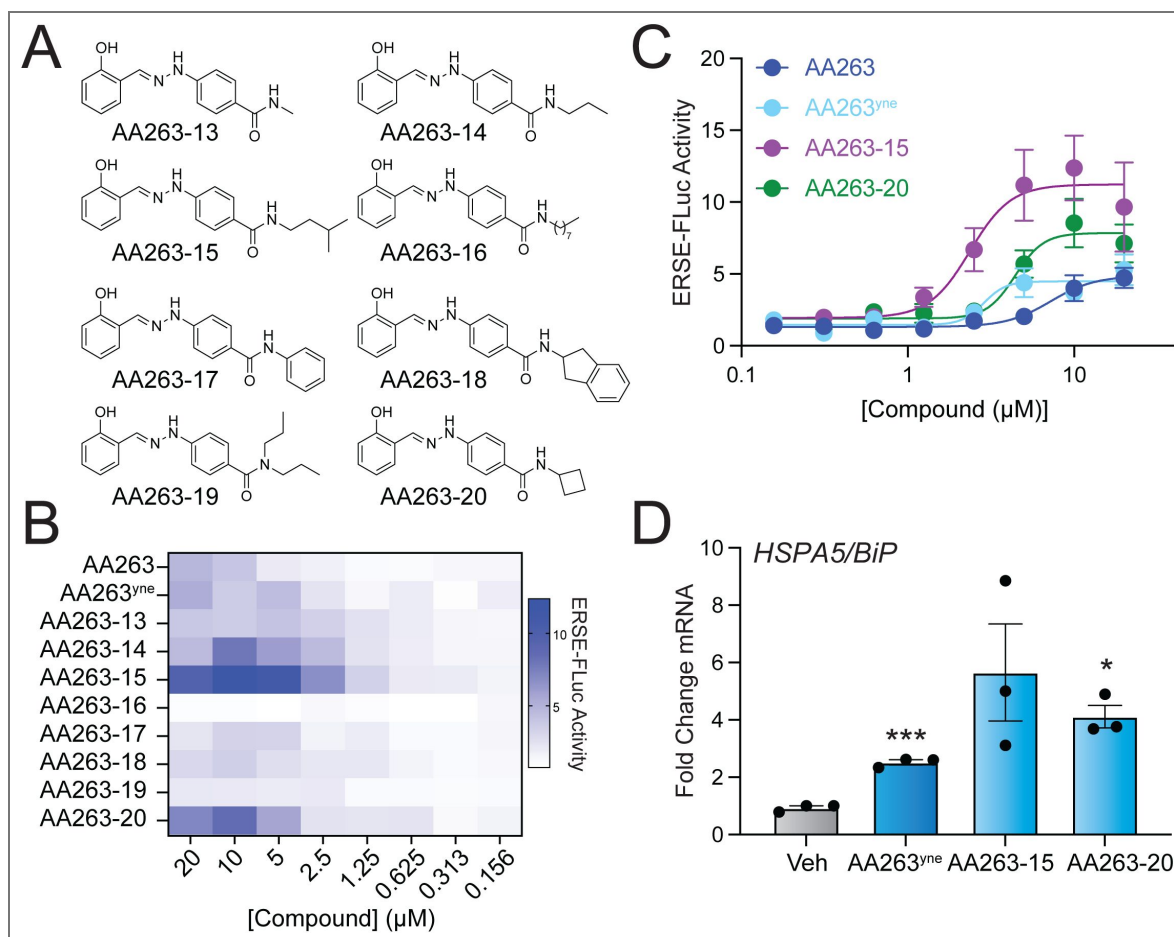


Figure 3. Diversification of the AA263 B-Ring affords improved AA263 analogs.

A. Structures of AA263 analogs. **B.** Heat map showing activation of the ERSE-FLuc ATF6 reporter in HEK293T cells treated for 18 h with the indicated dose of compound. **C.** Activation of the ERSE-FLuc ATF6 reporter in HEK293T cells treated for 18 h with the indicated dose of compound. Error bars show SEM for n=6 replicates. **D.** Expression, measured by qPCR, of the ATF6 target gene *BiP* in HEK293T cells treated with indicated AA263 analog (10 μ M) for 6 h. Error bars show SEM for n=3 independent biological replicates. *p<0.05, ***p<0.001 for one-way ANOVA.

(NE) inhibitor activity of secreted AAT in HUH7.5^{AATNull} cells transiently expressing mutant Z variant AAT.³³ Here, we compared the potential for AA263 and improved analogs of AA263 including AA263^{yne} and AA263-20 to mitigate polymer accumulation and reduced activity of the disease-associated Z variant of AAT stably expressed in liver-derived Huh7 cells (Huh7.5Z). Initially, we confirmed that AA263, AA263^{yne}, and AA263-20 induced expression of the ATF6 target gene *BiP* in these cells (Fig. S4). As reported previously, we found that AA263 reduced intracellular and secreted AAT-Z polymers, as measured by ELISA (Fig. 4A,B). However, we did not observe AA263-dependent enhancement of AAT NE inhibition activity in conditioned media, as measured by neutrophil elastase substrate-based fluorogenic assay (Fig. 4C).⁵⁶ In contrast, both AA263^{yne} and AA263-20 both reduced intracellular and secreted AAT polymers and enhanced AAT-Z NE inhibition in conditioned media (Fig. 4A-C). This indicates that AA263^{yne} and AA263-20 both show enhanced potential to rescue AAT-Z secretory proteostasis, as compared to AA263.

Optimized AA263 analogs rescue the surface trafficking and function of an epilepsy-prone GABA_A receptor variant

We next sought to investigate the potential of prioritized AA263 analogs for modulating the expression and functional activity of destabilized gamma-aminobutyric acid type A (GABA_A) receptors. GABA_A receptors are pentameric proteins assembled in the ER from eight subunit classes, with the most common composition being 2 α 1 subunits, 2 β 2/ β 3 subunits, and 1 γ 2 subunit.^{57,58} Various variants in these major subunits have been reported to impair the trafficking of GABA_A receptors to the plasma membrane, ultimately leading to a significant decrease in their function.^{59,60} As such, GABA_A variants have been widely associated with the onset and pathogenesis of numerous neurological disorders, including genetic epilepsy.^{11,59,61} It has previously been reported that enhancement of ER proteostasis can stabilize GABA_A variants and restore their surface trafficking as well as their function.^{34,62} Thus, we aimed to determine if our optimized AA263 analogs could similarly rescue the surface expression and activity of a known trafficking-deficient γ 2(R177G) GABA_A variant, associated with complex febrile seizures.⁶³

We treated HEK293T cells stably expressing α 1 β 2 γ 2(R177G) pathogenic receptors with each AA263 analog. While both AA263^{yne} and AA263-5 increased total protein levels of γ 2(R177G) (Fig. S5A), only AA263^{yne} could increase the surface expression of α 1 β 2 γ 2(R177G) GABA_A receptors transiently transfected in HEK293T cells (Fig. 5A). Thus, the optimized compound AA263^{yne} effectively rescues the trafficking of this GABA_A variant to the cell surface. Importantly, the increased total protein levels of γ 2(R177G) afforded by treatment with AA263^{yne} could be partially attenuated by co-treatment with the selective ATF6 inhibitor Ceapin-A7 (CP7)^{40–42}, indicating that compound-dependent ATF6 activation contributes to this observed increase (Fig. S5B). Next, we used cycloheximide (CHX) chase assays to determine the impact of AA263^{yne} on the turnover of the γ 2(R177G) variant. Compared with WT receptors, we found that the pathogenic γ 2(R177G) variant displayed a faster turnover, which was slowed down by treatment with AA263^{yne} to a similar rate compared to GABA_A WT (Fig. S5C). In addition to AA263^{yne}, both AA263-15 and AA263-20 increased the total expression levels of γ 2(R177G) subunit in HEK293T cells treated with either compound (Fig. S5D). Further, treatment with these compounds increased surface expression levels of the γ 2(R177G) subunit (Fig. 5B). Together, these results show that AA263^{yne} and optimized analogs such as AA263-20 can efficiently correct the misfolding of an epilepsy-associated GABA_A variant and rescue its trafficking to the plasma membrane.

Finally, to determine if compound-dependent increases of α 1 β 2 γ 2(R177G) receptor surface expression also translate into a rescued function of the receptor, we used whole-cell patch clamp in HEK293T cells transiently transfected with either WT GABA_A receptor or the α 1 β 2 γ 2(R177G) GABA_A variant and recorded evoked inhibitory postsynaptic currents (eIPSCs) by a 4-second application of 10 mM GABA. Consistent with the literature we first showed that cells expressing the epilepsy-associated γ 2(R177G) variant displayed eIPSCs of significantly smaller amplitude than cells expressing WT GABA_A receptors (Fig. 5C,D).⁶⁴ In addition, after normalizing each cell's amplitude by its capacitance, γ 2(R177G) GABA_A-containing cells had a lower peak current density when compared with WT GABA_A-containing cells (Fig. 5C,E). Importantly, treatment with

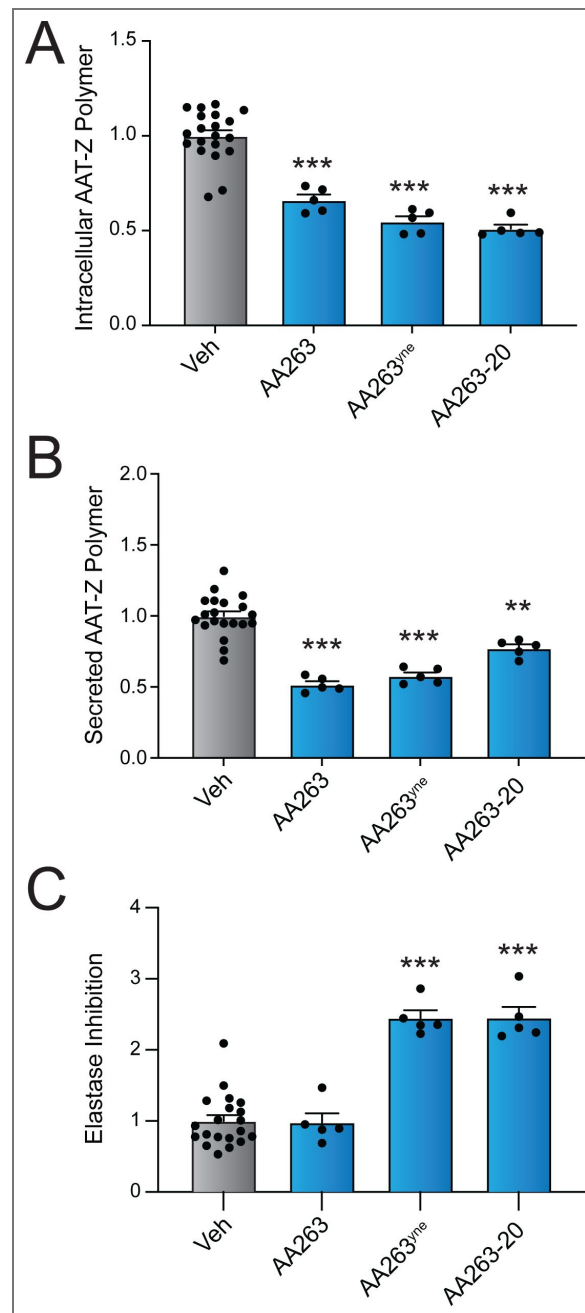


Figure 4. AA263 analogs improve secretory proteostasis for the disease-associated AAT-Z variant.

A-C. Intracellular AAT-Z polymer levels (**A**), extracellular AAT-Z polymer levels in conditioned media (**B**), and elastase inhibition activity of AAT-Z in conditioned media (**C**) from Huh7.5Z cells treated for 24 h with AA263 (10 μ M), AA263^{yne} (10 μ M), or AA263-20 (10 μ M). Error bars show SEM for $n > 5$ replicates. Data are shown normalized to vehicle-treated cells. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$ for one-way ANOVA compared to vehicle-treated cells.

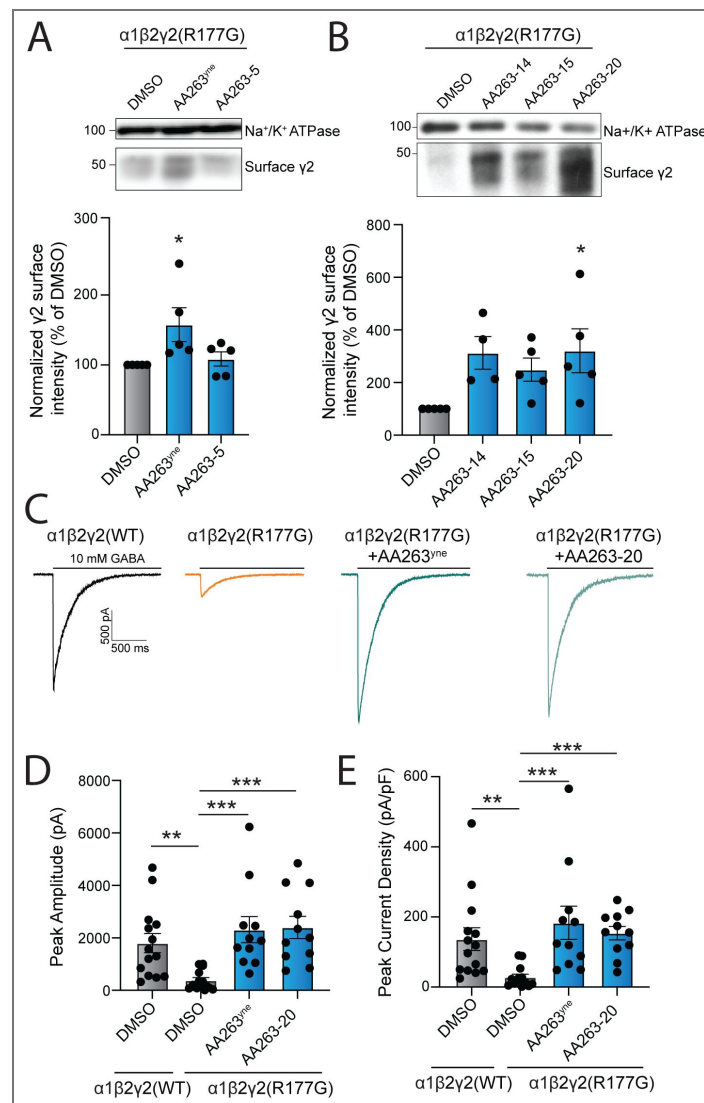


Figure 5. Enhanced AA263 analogs promote the trafficking and plasma membrane activity of destabilized, disease-associated GABA_A receptors.

A. Representative immunoblot (above) and quantification (below) of surface biotinylated $\gamma 2$ in HEK293T cells stably expressing $\alpha 1\beta 2\gamma 2(R177G)$ GABA_A receptors treated for 24 h with 10 μM AA263^{yne} or AA263-5. Na⁺/K⁺ ATPase serves as a loading control. **B.** Representative immunoblot showing surface $\gamma 2$ expression in HEK293T cells transiently transfected with $\alpha 1\beta 2\gamma 2(R177G)$ receptors and treated with indicated AA263 analogs (10 μM , 24hrs. Na⁺/K⁺ ATPase serves as a loading control. **C.** Representative eIPSC traces for $\alpha 1\beta 2\gamma 2(WT)$ GABA_A receptors and, $\alpha 1\beta 2\gamma 2(R177G)$ GABA_A receptors treated for 24 h with DMSO, AA263^{yne} (10 μM), or AA263-20 (10 μM). 10 mM GABA (saturating condition) was applied to the recorded cells to evoke currents. **D,E.** Histograms showing changes in eIPSC peak amplitude (D) and peak current density (E) for the indicated groups. Band intensities were quantified using ImageJ software, normalized to the DMSO control condition. Each data point is reported as mean $\pm$ SEM. One-way ANOVA followed by post-hoc Dunnett's test was used for statistical analysis for A and B. Kruskal-Wallis test followed by post-hoc Dunn's test was used for statistical analysis for D and E. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

AA263^{yne} or AA263-20 in HEK293T cells transiently expressing $\alpha 1\beta 2\gamma 2$ (R177G) GABA_A receptors restored both eIPSC peak amplitude and peak current density to levels nearly identical to those observed in cells expressing WT GABA_A receptors (**Fig. 5C-E**). Collectively, these findings establish AA263^{yne} and AA263-20 as ER proteostasis regulators that not only restore surface trafficking of disease-associated GABA_A receptor variants, but also rescue their functional activity at the cell surface.

Discussion

Herein, we define the molecular basis for AA263-dependent ER proteostasis remodeling. We show that AA263 covalently targets a subset of ER-localized PDIs, suggesting that this compound, like AA147, promotes activation of the adaptive ATF6 signaling arm of the UPR through a mechanism involving PDI modification. Intriguingly, unlike AA147, structure-activity relationship studies showed that AA263 activity could be enhanced by altering the putative B-ring of this compound. This identified two analogs, AA263-5 and AA263^{yne}, that exhibited enhanced selectivity and efficacy for ATF6 activation. Further optimization of the amide moiety of AA263^{yne} afforded prioritized analogs including AA263-20. We then showed AA263^{yne} and this prioritized analog exhibited enhanced activity relative to the parent compound AA263 in correcting ER proteostasis in cellular models of alpha-1- antitrypsin deficiency and GABA_A receptor channelopathies. These studies demonstrate the potential to enhance ER proteostasis remodeling activity with compound AA263 to correct proteostasis deficits in etiologically-diverse protein misfolding diseases.

Unlike other ER proteostasis activators, such as AA147 and AA132, AA263 does not possess the 2-amino-*p*-cresol moiety required for the enzymatic activation of these other compounds to a reactive quinone methide. Despite this, we show that AA263 functions through the covalent targeting of ER-localized PDIs. Salicylaldehyde N-acylhydrazones can generate transient quinone-methide type species driven by tautomerization.⁶⁵ This suggests that AA263 tautomerization to a quinone methide may underlie this covalent PDI targeting. Consistent with this, we show that AA263 can covalently modify proteins *ex vivo* in cellular lysates. However, a key question is how does AA263 show selectivity for ER PDIs independent of the localized metabolic activation afforded by other compounds such as AA147. One potential explanation is that the AA263 electrophilic species may possess differential reactivity towards a subset of proteinogenic thiols such as those in PDIs versus glutathione.^{66,67} Regardless, these results, when combined with our analyses for AA147 and AA132⁴⁹, demonstrate how the diverse covalent engagement profiles of subtly different electrophilic species against a subset of ER proteins can elicit a spectrum of transcriptional outcomes.

Our findings that AA263 engages similar PDI family members and ER proteins as AA147 further suggesting the importance of this family of proteins for adjusting ER proteostasis through mechanisms including activation of ER stress responsive signaling pathways.^{46,68–70} We previously showed that the extent of PDI labeling directly impacts the ability for this class of compounds to activate ATF6, with more modest levels leading preferential ATF6 activation while higher levels of labeling activating all three arms of the UPR.⁴⁹ Interestingly, our results indicate that AA263 maintains preferential selectivity for ATF6 activation despite modifying an intermediate population of PDIs greater than those observed for the selective ATF6 activator AA147, but less than those observed for the global UPR activator AA132. This suggests that higher levels of PDI engagement than those observed with AA147 are still accessible without compromising the ability for these compounds to preferentially activate ATF6.

Intriguingly, like AA147 and AA132, we show that the AA263 scaffold is amenable to modifications on the putative non-reactive B-ring. However, unlike these other compounds, medicinal chemistry efforts focused on improving AA263 activity identified analogs with improved potency and transcriptional selectivity for ATF6 activation relative to the parent compound. The improvements in transcriptional selectivity of the compounds may be attributed to the removal of the nitro group from the parent compound AA263. Nitroarene-containing compounds have been shown to affect mitochondrial function and also lead to ROS production through nitroarene metabolism.^{71,72} Thus, the crosstalk between ISR signaling and the mitochondria may explain the higher levels of

PERK/ISR signaling with AA263 relative to our prioritized analogs.^{73–75} Additionally, we do observe some basal CHOP signaling with inactive AA263 analogs, exemplified by AA263-2, perhaps suggesting some inherent promiscuity of the phenylhydrazone scaffold. As a result, the identity of the B-ring electron- withdrawing group may improve the spectrum of target protein engagement in a positive manner to boost selectivity for ATF6 activation.⁶⁶

It is important to note that both 2-hydroxyphenylhydrazones and 4-hydroxyphenylhydrazones have been classified as pan-assay interference compounds (PAINS), due, in part, to their reactivity towards biological nucleophiles and metal chelation properties.^{76,77 76,78} However, our observation of SAR trends connecting scaffold modifications to magnitude of pathway activation indicates that this particular compound activity is perhaps independent of PAINS activity. Further, many of our negative analogs possess a functional A-ring, and thus would still be expected to possess covalent reactivity if PAINS behavior mediated the activation of stress-responsive signaling.⁷⁹ Thus any nonspecific covalent modification, if it exists, seems to be functionally silent in this assay. Moreover, we found that these compounds showed remarkable specificity for ER-localized protein engagement, further indicating that AA263 and prioritized analogs possess other potential properties that drive specificity towards the ER, which we are continuing to pursue.

Optimized AA263 analogs identified herein have enhanced potential to correct ER proteostasis in two distinct cellular models of protein misfolding diseases – AAT deficiency and GABA_A receptor trafficking. We show that AA263 analogs reduce both intracellular and extracellular accumulation of AAT-Z aggregates and improve activity of the secreted protein, correcting both aspects of disease pathology. Further, we demonstrate that our optimized AA263 analogs enhanced trafficking, surface expression, and membrane activity of a pathogenic GABA_A receptor variant associated with genetic epilepsy, restoring is functional activity at the plasma membrane. AA263 and its related analogs can influence ER proteostasis in these models through different mechanisms including ATF6-dependent remodeling of ER proteostasis and direct alterations to the activity of specific PDIs. Consistent with this, we show that pharmacologic inhibition of ATF6 only partially blocks increases of $\gamma 2(R177G)$ afforded by treatment with AA263^{yne}, highlighting the benefit for targeting multiple aspects of ER proteostasis to enhance ER proteostasis of this disease-relevant GABA_A variant. While additional studies are required to further deconvolute the relative contributions of these two mechanisms on the protection afforded by our optimized compounds, our results demonstrate the potential for these compounds to enhance ER proteostasis in the context of different protein misfolding diseases. As we and others continue expanding on this class of ER proteostasis regulators, we will further demonstrate the therapeutic utility of these compounds in cellular and in vivo models and identify next-generation compounds with improved potential for translation to correct ER proteostasis defects in etiologically-diverse diseases.

Materials and methods

Reagents & Plasmids

Cycloheximide (cat #01810) and γ -aminobutyric acid (#A2129) were obtained from Sigma Aldrich. Human neutrophil elastase (Cat # IHUELASD100UG) was obtained from Innovation Research. MG-132 (#A2585) was obtained from ApexBio. Thapsigargin was purchased from Sigma-Aldrich. AA263 analogs and AA263^{yne} analogs were synthesized as described in *Supplemental Materials and Methods* and used as 10 mM DMSO stocks. The pCMV6 plasmids containing human GABA_A receptor $\alpha 1$ (Uniprot no. P14867-1), $\beta 2$ (isoform 2, Uniprot no. P47870-1), $\gamma 2$ (isoform 2, Uniprot no. P18507-2) subunits, and pCMV6 Entry Vector plasmid (pCMV6-EV) were obtained from Origene. The human GABA_A receptor $\gamma 2$ subunit missense mutation R177G was constructed using QuikChange II site-directed mutagenesis Kit (Agilent Genomics). All cDNA sequences were confirmed by DNA sequencing.

Antibodies

The rabbit anti-GABA_AR-γ2 polyclonal antibody (#AB5559) was obtained from Millipore. The rabbit monoclonal anti-Na⁺/K⁺-ATPase (#ab76020) antibody, was obtained from Abcam. The rhodamine anti-actin primary antibody (#12004163) was obtained from BioRad. Mouse anti-human AAT monoclonal antibody 2C1 was obtained from Hycult Biotech (Cat #HM2289). Mouse anti-human AAT monomer-specific monoclonal antibody 16F8 was a kind gift from the Balch Lab. The secondary goat anti rabbit antibody (#A27036) and goat anti mouse antibody (#A28177) used for western blot were obtained from Invitrogen. The secondary goat anti-mouse HRP antibody used for the AAT Elisa experiment was obtained from Thermo-Fischer scientific (Cat # 32230).

Cell Culture

HEK293T-Rex (ATCC), HEK293T (ATCC), and MEF (ATCC) cells were cultured in high-glucose Dulbecco's Modified Eagle's Medium (DMEM) supplemented with glutamine, penicillin/streptomycin and 10% fetal bovine serum (FBS). Cells were routinely tested for mycoplasma every 6 months. We did not further authenticate the cell lines. All cells were cultured under typical tissue culture conditions (37 °C, 5% CO₂).

Generation of Stable HEK293T α1β2γ2(R177G) Cell Line

HEK293T cells were grown in 10cm dishes and allowed to reach ~70% confluency before transient transfection using TransIT-2020 (Mirus Bio #MIR 5406, Madison, WI) according to the manufacturer's instruction. Stable cell lines for α1β2γ2(R177G) were generated using the antibiotic G-418 selection method. Briefly, cells were transfected with α1β2γ2(R177G) (0.25μg:0.25μg:0.5μg) plasmids, and then selected with 0.8 mg/mL G-418 (Fisher # 50-153-2785) for 7 days. Cells were diluted and passed to 96-well plates to ensure single cell distribution per well. Cells expanded from monoclonal cells were assessed for the expression of α1, β2, and γ2 by western blot analysis. Positive monoclonal cells were selected for experimentations.

Measurement of UPR activity using luciferase reporters

HEK293T-Rex cells expressing the ERSE.FLuc⁸⁰, XBP1s.RLuc⁸¹, or ATF4.FLuc reporter were plated at 80 μL/well from suspensions of 250,000 cells/mL in white clear-bottom 96-well plates (Corning) and incubated at 37°C overnight. The following day, cells were treated with 20 μL of compound-containing media to give final concentration as described before incubating for 18 hr at 37°C. The plates were equilibrated to room temperature, then either 100 μL of Firefly luciferase assay reagent-1 (ERSE.FLuc and ATF4.FLuc) or Renilla luciferase assay reagent-1 (XBP1s.RLuc) (Targeting Systems) were added to each well. Samples were dark adapted for 10 min to stabilize signals. Luminescence was then measured in an Infinite F200 PRO plate reader (Tecan) and corrected for background signal (integration time 500 ms). All measurements were performed in biologic triplicate.

Quantitative RT-PCR

The relative mRNA expression levels of target genes were measured using quantitative RT-PCR. Cells were treated as described at 37°C, harvested by trypsinization, washed with Dulbecco's phosphate-buffered saline (GIBCO), and then RNA was extracted using the QuickRNA Miniprep Kit (Zymo). qPCR reactions were performed on cDNA prepared from 500 ng of total cellular RNA using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). PowerSYBR Green PCR Master Mix (Applied Biosystems), cDNA, and appropriate primers purchased from Integrated DNA Technologies (see Table below) were used for amplifications (6 min at 95°C, then 45 cycles of 10 s at 95°C, 30 s at 60°C) in an ABI 7900HT Fast Real Time PCR machine. Primer integrity was assessed by a thermal melt to confirm homogeneity and the absence of primer dimers.

Transcripts were normalized to the housekeeping genes RPLP2 and all measurements were performed in biological triplicate. Data were analyzed using the RQ Manager and DataAssist 2.0 software (ABI). qPCR data are reported as mean ± standard deviation plotted using Prism

GraphPad.

Sequences of Primers for qPCR

GENE	FORWARD PRIMER	REVERSE PRIMER
HSPA5 (BIP)	GCCTGTATTTCTAGACCTGCC	TTCATCTTGCCAGCCAGTTG
ERDJ4	GGAAGGAGGAGCGCTAGGTC	ATCCTGCACCCTCCGACTAC
CHOP	ACCAAGGGAGAACCAGGAAACG	TCACCATTCCGGTCAATCAGAGC
RPLP2	CGTCGCCTCCTACCTGCT	CATTGAGCTCACTGATAACCTTG

SDS-PAGE In-Gel Fluorescence Scanning

Indicated cell line (250,000 cells/well) was treated with indicated compound or combination of compounds in six- well plates at 10 μ M for six hours. Cells were lysed in radioimmunoprecipitation assay (RIPA) buffer (150 mM NaCl, 50 mM Tris pH 7.5, 1% Triton X-100, 0.5% sodium deoxycholate, and 0.1% SDS) supplemented with fresh protease inhibitor cocktail (Roche, Indianapolis, IN) and centrifuged for 10 min at 16000 $\times$ g following a 30-minute incubation. Protein concentration of supernatant determined by BCA assay (Thermo Fisher) and normalized to give 42.5 μ L at 2.35 mg/mL (100 μ g/total protein). 7.5 μ L ‘click chemistry master mix’ was added to each sample to give final concentrations of 100 μ M of Cy5-azide (Click Chemistry Tools, Scottsdale, AZ), 800 μ M copper (II) sulfate, 1.6 mM BTAA ligand (2-(4-((bis((1-tert-butyl-1H-1,2,3-triazol-4-yl)methyl)amino)methyl)-1H-1,2,3-triazol-1-yl)acetic acid) (Albert Einstein College), and 5 mM sodium ascorbate. Reaction incubated at 30° C for 1 hour while shaking before CHCl_3 /MeOH protein precipitation. Dried protein was redissolved in 20 μ L 1X SDS loading buffer and 25 μ g was loaded on gel for SDS-PAGE in-gel fluorescence scanning and subsequent visualization using an Odyssey Infrared Imaging System (Li-Cor Biosciences) or BioRad gel imager.

Comparative AA132 vs. AA263 Chemoproteomic Experiments

HEK293T cells in 10 cm plates at 80-90% confluency were treated for 6 h with vehicle (0.1% DMSO), AA132yne (10 μ M), or AA263-5 (10 μ M) at 37 °C. The cells were washed with PBS before harvesting with trypsin, pelleting (500 g, 5 min), and washing with PBS (1 mL). Cells pellets were resuspended in radioimmunoprecipitation assay (RIPA) buffer before sonication with a probe tip sonicator for cell lysis (15 sec, 3 sec on/2 off, 30% amplitude). For each sample, 1 g lysate (500 μ L) were reacted with click reagents to give final concentrations as follows: 100 μ M of diazo biotin-azide (Click Chemistry Tools, Scottsdale, AZ), 800 μ M copper (II) sulfate, 1.6 mM BTAA ligand (2-(4-((bis((1-tert-butyl-1H-1,2,3-triazol-4-yl)methyl)amino)methyl)-1H-1,2,3-triazol-1-yl)acetic acid) (Albert Einstein College), and 5 mM sodium ascorbate. The reaction was placed on a shaker at 1000 rpm at 30 °C for 90 min. The reaction was quenched with the sequential addition of cold methanol (4x volume), chloroform (1x volume), and DPBS (4x volume) to precipitate proteins. Proteins were pelleted by centrifugation (4,700 g, 10 min, 4 °C). The supernatant was discarded, and the pellets dried under air for 5 min. Protein pellets were resuspended in 6M urea in PBS (500 μ L) with brief sonication. 50 μ L of high-capacity streptavidin beads were washed with PBS and mixed with the protein solution in 6 mL of phosphate-buffered saline (PBS). This suspension was placed on a rotator or a shaker and agitated for 2 h. The beads were centrifuged and washed 5 times with PBS and 1% SDS. The protein was eluted from the beads by two washes of 50 mM sodium dithionite in 1% SDS for 1 h and then precipitated by chloroform/methanol precipitation as described above. 50 μ L of freshly made 1:1 mixture 200 mM TCEP-HCl in DPBS and 600 mM K_2CO_3 in DPBS was added to each sample before incubation at 37 °C for 30 minutes while shaking. Alkylation of reduced

thiols was achieved by addition of 70 μ L freshly prepared 400 mM iodoacetamide in DPBS and incubation at room temperature while protected from light. The reaction was quenched by adding 130 μ L of 10% SDS in DPBS and then diluted to approximately 0.2% SDS via DPBS (5.5 mL) and incubated with preequilibrated streptavidin agarose beads (3x1 mL PBS wash). The samples were rotated at room temperature for 1.5 hours, centrifuged at 2000 rpm for 2 minutes, and then washed sequentially with 5 mL 0.2% SDS in DPBS, 5 mL DPBS, and 5 mL 100 mM TEAB (Thermo Cat #90114) pH 8.5 to remove non-binding proteins. The beads were transferred to low-bind 1.5 mL Eppendorf tubes and the bound proteins digested overnight at 37 °C in 200 μ L 100 mM TEAB containing 2 μ g sequencing grade porcine trypsin, 1 mM CaCl₂, and 0.01% ProteaseMax (Promega Cat #V2071). The beads were centrifuged at 2000 rpm for 5 minutes to separate the beads from the supernatant. 200 μ L supernatant was transferred to a new tube using a gel-loading tip, and the beads were washed with 100 μ L TEAB buffer. The beads were centrifuged at 2000 rpm for 5 minutes and the supernatant combined with the previous. 120 μ L acetonitrile was added to each supernatant sample before addition of 80 μ L (200 μ g) of TMT 10 plex (Thermo Scientific, cat #90110) reconstituted in acetonitrile. The samples were incubated at room temperature for 1 hour and vortexed occasionally. 7 μ L of freshly prepared 5% hydroxylamine in water was added to each sample to quench the reaction, vortexed, and incubated for 15 minutes before quenching with addition of 5 μ L MS-grade formic acid. The samples were then vacuum centrifuged to dryness. The samples were combined by redissolving the contents of one tube in 200 μ L 0.1 % trifluoroacetic acid solution in water and sequential transfer to the respective multiplexed experiment until all samples were redissolved. This stepwise process was repeated with an additional 100 μ L 0.1 % TFA solution for a final volume of 300 μ L. The pooled samples were fractionated using the Pierce high pH Reversed-Phase Fractionation Kit (Thermo Fisher Scientific 84868) according to manufacturer's instructions. The peptide fractions were eluted from the spin column with solutions of 0.1% triethylamine containing an increasing concentration of MeCN (5 - 95% MeCN; 8 fractions). Samples were dried via vacuum centrifugation, reconstituted in 50 μ L 0.1% formic acid, and stored at -80 °C until ready for mass spectrometry analysis.

LC-MS/MS analysis was performed using a Q Exactive mass spectrometer equipped with an EASY nLC 1000 (Thermo Fisher). The digest was injected directly onto a 30 cm, 75 μ m ID column packed with BEH 1.7 μ m C18 resin (Waters). Samples were separated at a flow rate of 200 nL/min on a nLC 1000 (Thermo). Buffer A and B were 0.1% formic acid in water and acetonitrile, respectively. A gradient of 5–40% B over 110 min, an increase to 50% B over 10 min, an increase to 90% B over another 10 min and held at 90% B for a final 10 min of washing was used for 140 min total run time. The column was re-equilibrated with 20 μ L of buffer A prior to the injection of sample. Peptides were eluted directly from the tip of the column and nanosprayed directly into the mass spectrometer by application of 2.5 kV voltage at the back of the column. The Q Exactive was operated in a data- dependent mode. Eluted peptides were scanned from 400 to 1800 m/z with a resolution of 30,000 and the mass spectrometer in a data dependent acquisition mode. The top ten peaks for each full scan were fragmented by HCD using a normalized collision energy of 30%, a 100 ms activation time, a resolution of 7500, and scanned from 100 to 1800 m/z. Dynamic exclusion parameters were 1 repeat count, 30 ms repeat duration, 500 exclusion list size, 120 s exclusion duration, and exclusion width between 0.51 and 1.51. Peptide identification and protein quantification was performed using the Integrated Proteomics Pipeline Suite (IP2, Integrated Proteomics Applications, Inc., San Diego, CA) as described previously.

RNA-seq

Cells were lysed and total RNA collected using the Quick-RNA Miniprep kit from Zymo Research (R1055) according to manufacturer's instructions. RNA concentration was then quantified by NanoDrop. Whole transcriptome RNA was then prepared and sequenced by BGI Americas on the BGI Proprietary platform, which provided paired-end 50 bp reads at 20 million reads per sample. Each condition was performed in triplicate. RNAseq reads were aligned using DNASTar Lasergene SeqManPro to the Homo_sapiens-GRCh38.p7 human genome reference assembly, and assembly data were imported into ArrayStar 12.2 with QSeq (DNASTar Inc.) to quantify the gene expression levels and normalization to reads per kilobase per million. Differential expression analysis was

assessed using DESeq2 in R, which also calculated statistical significance calculations of treated cells compared to vehicle-treated cells using a standard negative binomial fit of the reads per kilobase per million data to generate fold-change quantifications. The complete RNAseq data is deposited in gene expression omnibus (GEO) as GSE309691.

Immunoblotting

Cells were grown in 6-well plates or 10-cm dishes and allowed to reach ~70% confluency before transient transfection with $\alpha 1:\beta 2:\gamma 2$ (WT) or $\alpha 1:\beta 2:\gamma 2$ (R177G) (0.25 μ g:0.25 μ g:0.5 μ g) plasmids using TransIT-2020 (# Mirus Bio, Madison, WI) according to the manufacturer's instruction. The mixture was incubated for 15-20 minutes at room temperature before being added to the cells. After indicated treatment, cells were harvested with Trypsin and lysed with pH 7.5 lysis buffer: (50 mM Tris, 150 mM NaCl, 2 mM *n*-Dodecyl- β -D-maltoside (DDM) (GoldBio, catalog #: DDM5)) (pH 7.5) supplemented with complete protease inhibitor cocktail (Roche #4693159001). Lysates were cleared by centrifugation (21,000 xg, 10 min, 4°C). Protein concentrations were then measured through microBCA assay (ThermoFisher Pierce #23235). Aliquots of cell lysates were loaded with 4x Laemmli buffer (Biorad #1610747, CA) with 10% 2-Mercaptoethanol (Sigma Aldrich #M3148, Saint Louis, MO) and separated in an 8% SD-PAGE gel. The β -actin serves as a total protein loading control, whereas Na^+/K^+ -ATPase serves as a plasma membrane protein loading control. Band intensity was quantified using Image-J software from the NIH. For both surface and total expression quantification, the protein level was first normalized to the loading control (β -actin or Na^+/K^+ -ATPase) and then to the vehicle control (DMSO or WT).

AAT-Z specific conformation (monomer or polymer) ELISA assay

Huh7.5Z cells stably expressing AAT-Z (E342K)⁸² were seeded in 96-well plates at a density of 2.5×10^4 cells per well. Cells were treated with compounds for 24 h before collecting the cell culture medium. Wells were aspirated of residual media & refilled with 80 μ L/well of lysis buffer (50 mM Tris, 150mM NaCl, 1% (v/v) Triton X-100, and 1X Halt™ protease inhibitor cocktail) and incubated on 4°C shaker for 1 h before collecting the resultant lysate. The previous day, clear flat bottom polystyrene high binding 96-well plates were pre-coated with goat anti-human AAT polyclonal antibody (A80-122A 1:1000 in coating buffer (28.6mM g Na₂CO₃, 71.4mM NaHCO₃, pH = 9.6)) at 4°C. Coated plates were washed 5x with PBST washing buffer (1X PBS + 0.05% (v/v) Tween-20), blocked for 2-4 h in washing buffer containing 5% dehydrated milk, and washed another 5x with to remove blocking agent just prior to collection of treated cell products. 20 μ L of culture medium or lysate from each treated well was dispensed into the antibody-coated plates and incubated overnight (8-12 h) at 4°C. Plates were washed 5x with PBST washing buffer before applying conformation specific primary antibodies for AAT monomer (16F8⁸³ 1:2000 in 1X PBS) or polymer (2C1⁸⁴ 1:1000 in 1X PBS) and incubating 2-4 h at room temperature. Plates were washed 5x with PBST washing buffer before applying the secondary HRP conjugated goat-anti-mouse antibody (1:5000 in 1xPBS) and incubating 2-4 h at room temperature. After washing a final 5x with PBST washing buffer, TMB substrate (Thermo Fisher Scientific, No. 34029) was added for 10 min before quenching the reaction with 2 M sulfuric acid. Endpoint ELISA signal (450nm absorbance) was quantified by BioTek Synergy H1 Hybrid plate reader. AAT-Z monomer and polymer levels in media and lysate following 24 h treatment with AA263 analogues were normalized to vehicle treated Huh7.5Z cells.

AAT-Z inhibitory activity (fluorogenic elastase substrate turnover) assay

Culture medium from treated Huh7.5Z cells was collected and added to pre-coated high binding 96-well plates as described above for overnight (8-12 h) incubation at 4°C. Plates were washed 5x with PBST washing buffer before dispensing a solution of porcine elastase (PPE, Promega, No. V1891) at 4 ng/well in reaction buffer (50mM Tris, pH 8.5) and incubating the plate for 1hr at 37°C. Then, 25 pmol/well of the fluorogenic elastase substrate (Z-Ala4)2Rh110 (Cayman Chemical, No. 11675) was added, and the plate incubated for 1.5 h at 37°C. The plate was read by a BioTek

Synergy H1 Hybrid plate reader at 485nm excitation and 525nm emission. Elastase inhibitory activity of secreted AAT-Z in media following 24 h treatment with AA263 analogues was normalized to vehicle treated Huh7.5Z cells.

Biotinylation of Cell Surface Proteins

HEK293T cells were plated in 6cm dishes for surface biotinylation assays. Cells were transfected with GABA_A receptor variants as indicated 48h prior to harvest. 24h post incubation with the indicated compounds, intact cells were washed with ice-cold Dulbecco's Phosphate Buffered Saline (DPBS, VWR #10128-844) and incubated with the membrane-impermeable biotinylation reagent Sulfo-NHS SS-Biotin (1mg/mL; ApexBio, #A8005) in DPBS containing 0.1mM CaCl₂ and 1mM MgCl₂ (DPBS + CM) for 30 min at 4°C to label surface membrane proteins. Cells were then incubated with 10mM Glycine twice at 4°C for 5 min to quench reaction. To block sulfhydryl groups, the cells were then incubated with 5mM N-ethyl-maleimide (NEM) in DPBS for 15 min at room temperature. Cells were then solubilized overnight at 4°C in lysis buffer containing 2mM DDM, 50mM Tris-HCL, 150mM NaCl, and 5mM ETA (pH 7.5) supplemented with Roche complete protease inhibitor cocktail (Roche #04,693,116,001) and 5mM NEM. To pellet cellular debris, the lysates were cleared by centrifugation (21,000 x g, 10 min, 4°C). The supernatant containing the biotinylated surface proteins was affinity-purified by incubating for 1h at 4°C with 40μL of immobilized neutravidin-conjugated agarose bead slurry (VWR #PI29201). The samples were then subjected to centrifugation (21,000 x g, 10 min, 4°C). The beads were washed three times with a buffer containing in mM: 50 Tris-HCL, 150 NaCl, and 5 EDTA (pH 7.5). Surface proteins were finally eluted from beads by vortexing for 30 min with 80μL of a buffer containing 2 x Laemmli sample buffer, 100 mM DTT, and 6 M urea (pH 6.8) for SDS-PAGE and Western blotting analysis.

Cycloheximide-Chase Assay

HEK293T cells transiently transfected with α1β2γ2(WT) GABA_A receptors or stably expressing α1β2γ2(R177G) GABA_A receptors were seeded in 6-well plates and incubated at 37 °C overnight. Cells were then treated with 10 μM compound for 24h prior the commencement of the experiment. To stop protein translation, cells were treated with 100 μg/mL cycloheximide (Enzo Life Sciences #ALX-380-269). Cells were then chased for the indicated times, harvested, and lysed for protein analysis.

Electrophysiological Recordings in Transfected Hek Cells by Whole-Cell Patch Clamp

GABA_A α1β2γ2 (WT or R177G) subunits were transfected 48h prior recordings at a subunit plasmid ratio of 0.5α:0.5β:1γ (total DNA was 0.25-0.5 μg) into HEK293T cells using TransIT-2020 reagent (Mirus Bio #MIR 5406, Madison, WI). The α1(WT) subunit was co-transfected with GFP, which acted as an expression marker. ATF6 activators at 10 μM were added to the cultures 24h post-transfection, and 24h prior to the commencement of the experiments.

All experiments were performed at room temperature (RT) in the whole-cell configuration obtained in voltage-clamp mode at a holding potential of -60 mV. Patch pipettes with a resistance of 3 to 5 MΩ were made from borosilicate glass (DWK Life Sciences, #34500-99) on a two-step micropipette puller (Narishige PP-830). The intracellular solution contained the following (in mM): 153 KCl, 1 MgCl₂, 5 (ethylene glycolbis(b-aminoethyl ether)-N,N,N₉,N₉-tetraacetic acid (EGTA), 10 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), and 5 Mg-ATP with pH adjusted to 7.3 and osmolarity at 310 mOsm. Cells were continuously perfused with an extracellular solution composed of (in mM): 142 NaCl, 8 KCl, 6 MgCl₂, 1 CaCl₂, 10 Glucose, and 10 HEPES with pH adjusted to 7.4 and osmolarity at 290 mOsm.

Each experiment constituted of one minute recording in extracellular solution with the application of 10 mM GABA (Sigma Aldrich #A2129, Saint Louis, MO) for 4 seconds every 6 seconds to the recorded cell via a theta glass (Siskiyou #15700000E) controlled by a high-speed piezo solution switcher (Siskiyou, MXPZT-300 series).

Evoked inhibitory post-synaptic currents (eIPSCs) were recorded using an Axopatch 200B amplifier and pClamp 10 software, filtered at 1 kHz and sampled at 20 kHz. The first action potential of each recording was analyzed for peak amplitude (pA) using Clampfit 11.2. Normalization of currents to each cell's capacitance (pF) was performed to allow for collection of current density data (pA/pF). Recordings with an access resistance $> 12 \text{ M}\Omega$ were not included in the analysis. For each experiment, GFP fluorescence was used to control for successful expression of the GABA_A receptor in the cells.

Quantification and Statistical Analysis

Statistical analyses were performed using GraphPad Prism 10 (GraphPad, San Diego, CA). All data were first tested for a Gaussian distribution using a Shapiro-Wilk test. The statistical significance between means was determined by either parametric or nonparametric analysis of variance followed by post-hoc comparisons (Dunn's, Dunnett) using GraphPad Prism 10. Statistical significance was set at a < 0.05 . Error bars in the graphs represent mean $\pm$ SEM.

Figure supplements

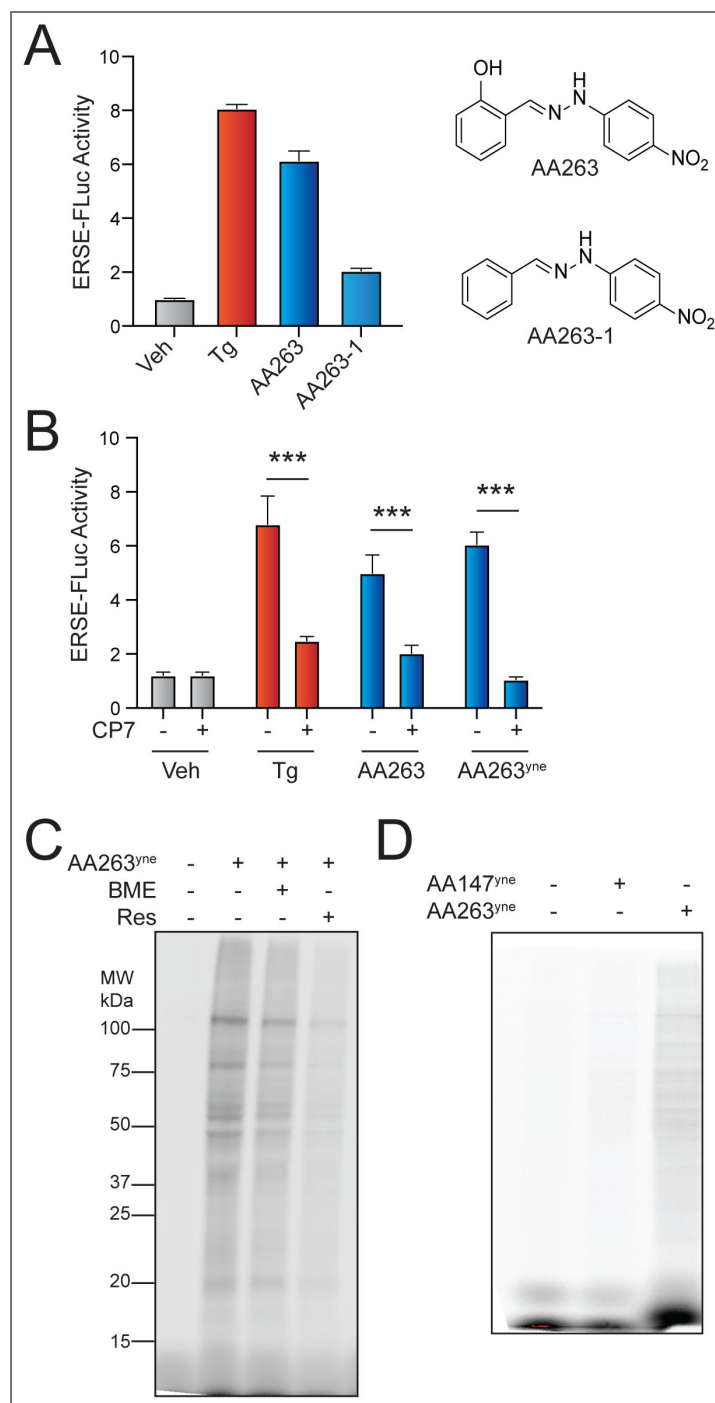


Figure S1 (Supplement to Figure 1). AA263 covalently modifies protein disulfide isomerase family members. A. Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated for 18 h with vehicle (0.1% DMSO), Tg (500 nM), AA263(10 μ M), or AA263-1 (10 μ M). Structures of AA263 and AA263-1 shown to the right. Error bars show SEM for $n > 3$ replicates. **B.** Activation of the ERSE.FLuc ATF6 reporter in HEK293T cells treated for 18 h with vehicle (0.1% DMSO), Tg (500 nM), AA263 (10 μ M), or AA263^{yne} (10 μ M), and/or Ceapin-A7 (CP7; 10 μ M). Error bars show SEM for $n = 3$ replicates. *** $p < 0.005$ for two-way ANOVA. **C.** Representative SDS-PAGE gel of Cy5-conjugated proteins from HEK293T cells treated for 4 h with vehicle (0.1% DMSO), AA263^{yne} (10 μ M), or the

combination of AA263^{yne} (10 μ M) and either resveratrol (10 μ M) or BME (55 μ M). **D.** Representative SDS-PAGE gel of Cy5-conjugated proteins from lysates prepared from HEK293T treated for 1h with vehicle, AA147^{yne}, or AA263^{yne} (5 μ M).

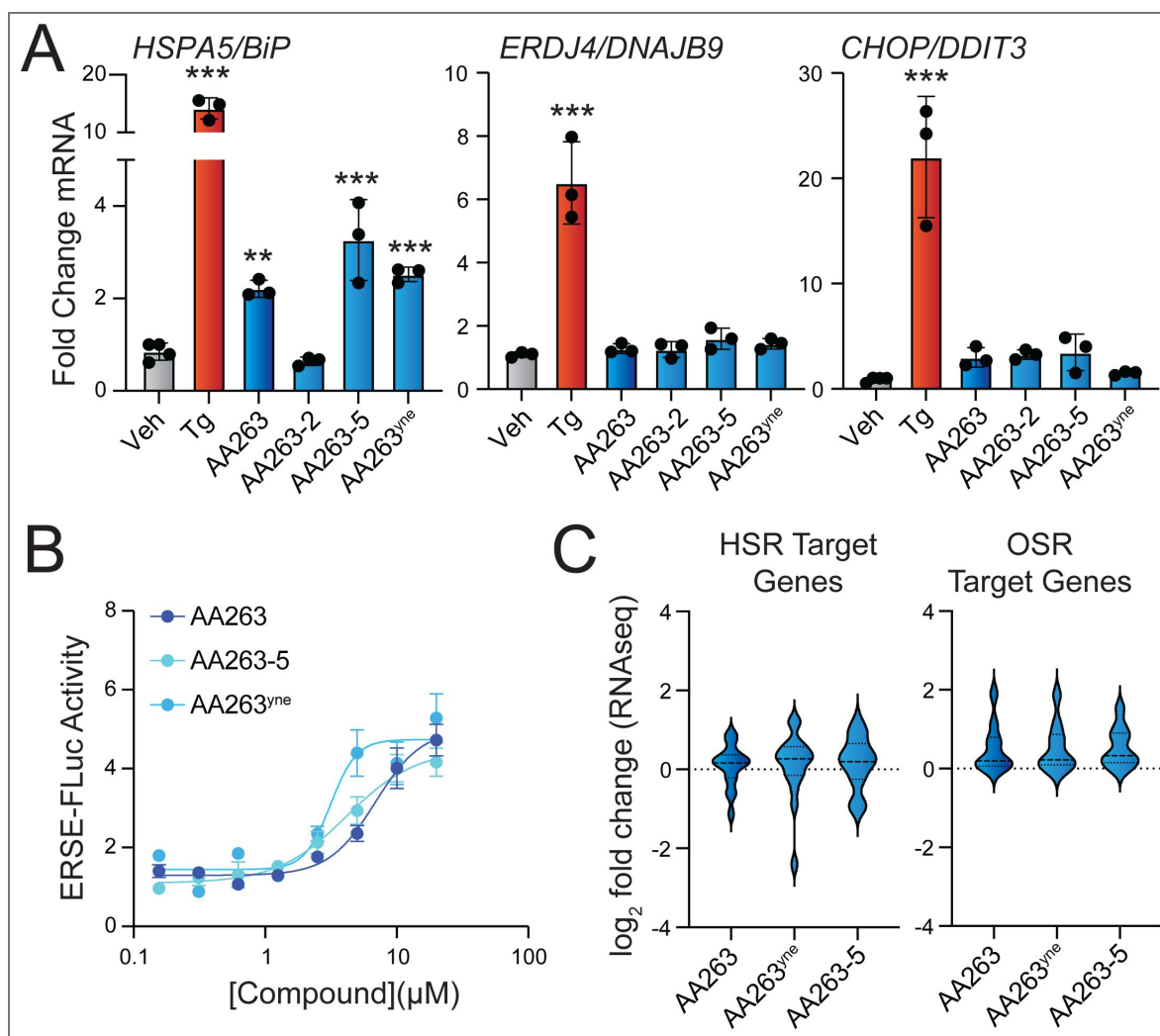


Figure S2 (Supplement to Figure 2). Identification of AA263 analogs that show enhanced ATF6 activation.

A. Expression, measured by qPCR, of the ATF6 target gene *BiP*, the IRE1/XBP1s target gene *DNAJB9*, and the PERK/ISR target gene *CHOP* in HEK293T cells treated for 6 h with the indicated compound (10 μ M). Error bars show SEM for $n = 3$ biological replicates. * $p < 0.05$, *** $p < 0.001$ for one-way ANOVA relative to vehicle-treated cells. **B.** Activation of the ERSE-FLuc ATF6 reporter in HEK293T cells treated for 18 h with the indicated concentration of AA263, AA263^{yne}, or AA263-5. Error bars show SEM for $n = 3$ replicates. The data for AA263 and AA263^{yne} is the same as that shown in Fig. 1E and are included for comparison. **C.** Expression, measured by RNAseq, of gene sets comprising target genes regulated downstream of the heat shock response (HSR, left) or the oxidative stress response (OSR, right) in HEK293 cells treated for 6 h with 10 μ M AA263, AA263^{yne}, or AA263-5. Full RNAseq data and genesets used in this analysis are shown in Table S1

Figure S3 (Supplement to Figure 3). Diversification of the AA263 B-Ring affords improved AA263 analogs.

A. Expression, measured by qPCR, of the IRE1/XBP1s target gene *DNAJB9* and the PERK/ISR target gene *CHOP* in HEK293T cells treated for 6 h with the indicated AA263 analog (10 μ M). Error bars show SEM for n=3 independent biological replicates. **B.** Representative SDS-Page gel of Cy5-conjugated proteins from HEK293T cells treated for 4 h with vehicle, AA263^{yne} (10 μ M) or cotreatment of AA263^{yne} (10 μ M) with the indicated analog (40 μ M).

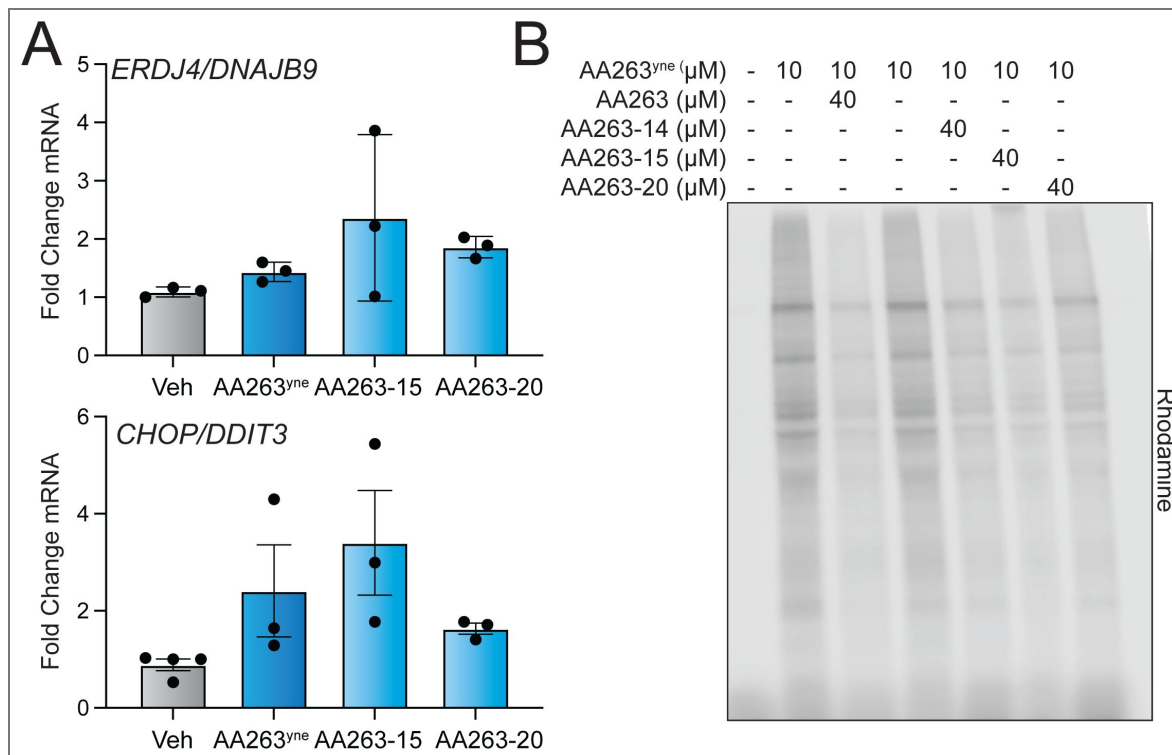
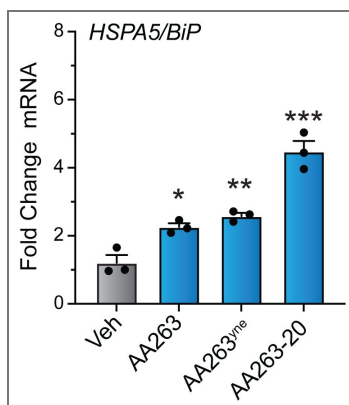


Figure S4 (Supplement to Figure 4). AA263 analogs improve secretory proteostasis for the disease-associated AAT-Z variant.

Expression, measured by qPCR, of the ATF6 target gene *BiP* in Huh7.5Z cells treated for 6 h with 10 μ M of AA263, AA263^{yne}, or AA263-20. Error bars show SEM for n=3 replicates. *p<0.05, **p<0.01, ***p<0.005 for one-way ANOVA compared to vehicle-treated cells are shown.



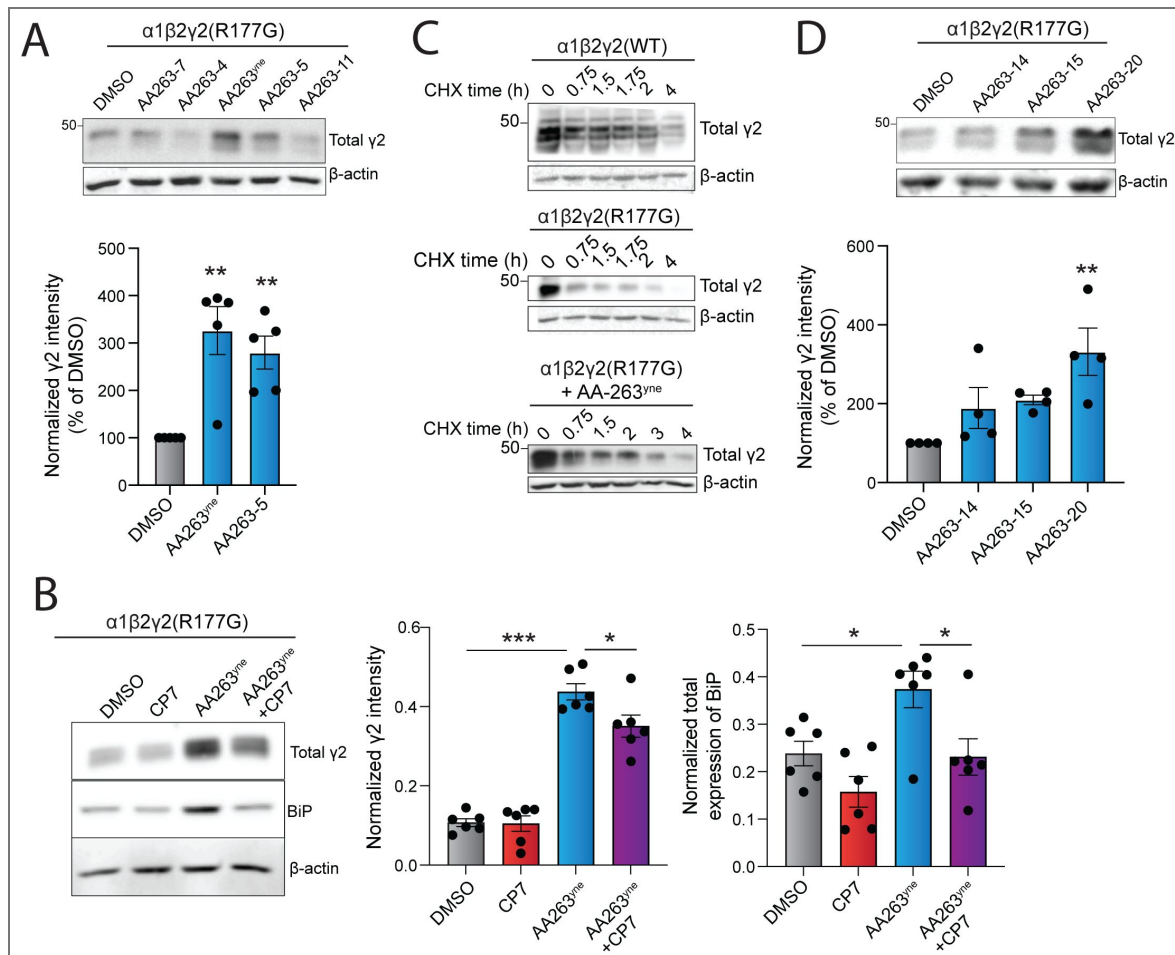


Figure S5 (Supplement to Figure 5). Enhanced AA263 analogs promote the trafficking and plasma membrane activity of destabilized, disease-associated GABA_A receptors.

A. Representative immunoblot (above) of HEK293T cells stably expressing $\alpha 1\beta 2\gamma 2(R177G)$ GABA_A receptors treated for 24h with the indicated AA263 analog (10 μ M). Quantification of band intensities (below) for cells treated with AA263^{yne} or AA263-5 is shown. **B.** Immunoblot and quantification of $\gamma 2$ and BiP in lysates prepared on HEK293T cells stably expressing $\alpha 1\beta 2\gamma 2(R177G)$ GABA_A receptors treated for 24h with AA263^{yne} (10 μ M) and/or Ceapin-A7 (CP7; 10 μ M). **C.** Immunoblot showing the intensity of $\gamma 2$ over time following cycloheximide (CHX)-chase application (0 to 4 hours, 100 μ g/mL) in HEK293T cells transiently transfected with $\alpha 1\beta 2\gamma 2(WT)$ receptors (top) and HEK293T cells stably expressing $\alpha 1\beta 2\gamma 2(R177G)$ receptor variant treated with vehicle (middle) or AA263^{yne} (bottom). **D.** Representative immunoblot showing total $\gamma 2$ expression in HEK293T cells stably expressing $\alpha 1\beta 2\gamma 2(R177G)$ receptor variant treated with indicated AA263^{yne} analogs (10 μ M, 24hrs). One-way ANOVA followed by post-hoc Dunnett's test was used for statistical analysis. * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.0001$.

Acknowledgements

We thank Leonard Yoon for help in analyzing the RNAseq experiments described in this manuscript. This work was supported by the National Institutes of Health (AG046495 to RLW and JWK; DK137470 to RLW; NS105789 to TM; HL169631 to WEB).

Additional files

Table S1 [↗](#) RNAseq analysis of HEK293 cells treated with AA263 (10 μ M), AA263^{yne} (10 μ M), or AA263-5 (10 μ M). There are four sheets in this Excel workbook. DESEQ outputs for RNAseq in HEK293 cells treated with 10 μ M AA263, AA263^{yne}, or AA263-5 for 6 h and a sheet showing the geneset profiling of different stress-responsive signaling pathways from these RNAseq data.

Table S2 [↗](#) GO analysis of RNAseq from HEK293 cells treated with AA263 (10 μ M), AA263^{yne} (10 μ M), or AA263-5 (10 μ M). There are three sheets in this Excel workbook showing the GO analysis results of RNAseq data from HEK293 cells treated for 6 h with 10 μ M AA263, AA263^{yne}, or AA263-5. GO analysis was performed on genes induced greater than 1.5-fold with an adjusted p value less than 0.05.

Figure 1 - Source data 1 [↗](#)

Figure 2 - Source data 1 [↗](#)

Figure 2/S2 - Source data 2 [↗](#)

Figure 2/S2 - Source data 3 [↗](#)

Figure 3 - Source data 1 [↗](#)

Figure 4 - Source data 1 [↗](#)

Figure 5 - Source data 1 [↗](#)

Figure S1 - Source data 1 [↗](#)

Figure S2 - Source data 1 [↗](#)

Figure S3 - Source data 1 [↗](#)

Figure S4 - Source data 1 [↗](#)

Figure S5 - Source data 1 [↗](#)

Additional information

Funding

Funder	Grant reference number	Author
HHS National Institutes of Health (NIH)	AG046495	Jeffery W KellyLuke Wiseman
HHS National Institutes of Health (NIH)	DK137470	Luke Wiseman
HHS National Institutes of Health (NIH)	NS105789	Tingwei Mu
HHS National Institutes of Health (NIH)	HL169631	William E Balch

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study builds off prior work that focused on the molecule AA147 and its role as an activator of the ATF6 arm of the unfolded protein response. In prior manuscripts, AA147 was shown to enter the ER, covalently modify a subset of protein disulfide isomerases (PDIs), and improve ER quality control for the disease-associated mutants of AAT and GABAA. Unsuccessful attempts to improve the potency of AA147 have led the authors to characterize a second hit from the screen in this study: the phenylhydrazone compound AA263. The focus of this study on enhancing biological activity of the AA147 molecule is compelling, and overcomes a hurdle of the prior AA147 drug that proved difficult to modify. The study successfully identifies PDIs as a shared cellular target of AA263 and its analogs. The authors infer, based on the similar target hits previously characterized for AA147, that PDI modification likely accounts for a mechanism of action for AA263.

Strengths:

The work establishes the ability to modify the AA263 molecule to create analogs with more potency and efficacy for ATF6 activation. The "next generation" analogs are able to enhance the levels of functional AAT and GABAA receptors in cellular models expressing the Z-variant of AAT or an epilepsy-associated variant of the GABAA receptor, outlining the therapeutic potential for this molecule and laying the foundation for future organism-based studies.

The authors are able to establish that like AA147, AA263 covalently targets ER PDIs. While it is a likely mechanism that AA263 works through the PDIs, the authors are careful to discuss that this is a potential mechanism that remains to be explicitly proven. The study provides the foundation for future work to further define a role for the PDIs in the actions of AA263.

<https://doi.org/10.7554/eLife.107000.2.sa3>

Reviewer #2 (Public review):

Modulating the UPR by pharmacological targeting of its sensors (or regulators) provides mostly uncharted opportunities in diseases associated with protein misfolding in the secretory pathway. Spearheaded by the Kelly and Wiseman labs, ATF6 modulators were developed in previous years that act on ER PDIs as regulators of ATF6. However, hurdles in their medicinal chemistry have hampered further developments. In this study, the authors provide evidence that the small molecule AA263 also targets and covalently modifies ER PDIs with the effect of activating ATF6. Importantly, AA263 turned out to be amenable to chemical optimization while maintaining its desired activity. Building on this, the authors show that AA263 derivatives can improve aggregation, trafficking and function of two disease-associated mutants of secretory pathway proteins. Together, this study provides compelling evidence for AA263 (and its derivatives) being interesting modulators of ER proteostasis. Mechanistic details of its mode of action will need more attention in future studies that can now build on this.

In detail, the authors provide strong evidence that AA263 covalently binds to ER PDIs, which will inhibit the protein disulfide isomerase activity. ER PDIs regulate ATF6, and thus their

finding provides a mechanistic interpretation of AA263 activating the UPR. It should be noted, however, that AA263 shows broad protein labeling (Fig. 1G) which may suggest additional targets, beyond the ones defined as MS hits in this study. Also, a further direct analysis of the IRE1 and PERK pathways (activated or not by AA263) may be an interesting future directions, as e.g. PDIA1, a target of AA263, directly regulates IRE1 (Yu et al., EMBOJ, 2020) and other PDIs also act on PERK and IRE1. The authors interpret modest activation of IRE1/PERK target genes (Fig. 2C) as an effect on target gene overlap, indeed the most likely explanation based on their selective analyses on IRE1 (ERdj4) and PERK (CHOP) downstream genes, but direct activation due to the targeting of their PDI regulators is also a possible explanation. Further key findings of this paper are the observed improvement of AAT behavior and GABAA trafficking and function. Further strength to the mechanistic conclusion that ATF6 activation causes this could be obtained by using ATF6 inhibitors/knockouts in the presence of AA263 (as the target PDIs may directly modulate behavior of AAT and/or GABAA). Along the same line, it also warrants further investigation in future studies why the different compounds, even if all were used at concentrations above their EC50, had different rescuing capacities on the clients.

Together, the study now provides a strong basis for such in-depth mechanistic analyses.

<https://doi.org/10.7554/eLife.107000.2.sa2>

Reviewer #3 (Public review):

Summary:

This study aims to develop and characterize phenylhydrazone-based small molecules that selectively activate the ATF6 arm of the unfolded protein response by covalently modifying a subset of ER-resident PDIs. The authors identify AA263 as a lead scaffold and optimize its structure to generate analogs with improved potency and ATF6 selectivity, notably AA263-20. These compounds are shown to restore proteostasis and functional expression of disease-associated misfolded proteins in cellular models involving both secretory (AAT-Z) and membrane (GABAA receptor) proteins. The findings provide valuable chemical tools for modulating ER proteostasis and may serve as promising leads for therapeutic development targeting protein misfolding diseases.

Strengths:

The study presents a well-defined chemical biology framework integrating proteomics, transcriptomics, and disease-relevant functional assays.

Identification and optimization of a new electrophilic scaffold (AA263) that selectively activates ATF6 represents a valuable advance in UPR-targeted pharmacology.

SAR studies are comprehensive and logically drive the development of more potent and selective analogs such as AA263-20.

Functional rescue is demonstrated in two mechanistically distinct disease models of protein misfolding-one involving a secretory protein and the other a membrane protein-underscoring the translational relevance of the approach.

Weaknesses:

ATF6 activation is primarily inferred from reporter assays and transcriptional profiling; direct biochemical evidence of ATF6 cleavage or nuclear translocation remains missing. However, the authors have added supporting data showing that co-treatment with the ATF6 inhibitor CP7 suppresses target gene induction, which partially strengthens the evidence for ATF6-dependent activity.

Although the proposed mechanism involving PDI modification and ATF6 activation is plausible, it is still not experimentally demonstrated and remains incompletely characterized.

In vivo validation is absent, and thus the pharmacological feasibility, selectivity, and bioavailability of these compounds in physiological systems remain untested.

Comments on revisions:

The authors have generally addressed my comments.

<https://doi.org/10.7554/eLife.107000.2.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Public review):

Summary:

This study builds off prior work that focused on the molecule AA147 and its role as an activator of the ATF6 arm of the unfolded protein response. In prior manuscripts, AA147 was shown to enter the ER, covalently modify a subset of protein disulfide isomerases (PDIs), and improve ER quality control for the disease-associated mutants of AAT and GABAA. Unsuccessful attempts to improve the potency of AA147 have led the authors to characterize a second hit from the screen in this study: the phenylhydrazone compound AA263. The focus of this study on enhancing the biological activity of the AA147 molecule is compelling, and overcomes a hurdle of the prior AA147 drug that proved difficult to modify. The study successfully identifies PDIs as a shared cellular target of AA263 and its analogs. The authors infer, based on the similar target hits previously characterized for AA147, that PDI modification accounts for a mechanism of action for AA263.

Strengths:

The authors are able to establish that, like AA147, AA263 covalently targets ER PDIs. The work establishes the ability to modify the AA263 molecule to create analogs with more potency and efficacy for ATF6 activation. The "next generation" analogs are able to enhance the levels of functional AAT and GABAA receptors in cellular models expressing the Z-variant of AAT or an epilepsy-associated variant of the GABAA receptor, outlining the therapeutic potential for this molecule and laying the foundation for future organism-based studies.

We thank the reviewer for the positive comments on our manuscript. We address the reviewers remaining comments on our work, as described below.

Weaknesses:

Arguably, the work does not fully support the statement provided in the abstract that the study "reveals a molecular mechanism for the activation of ATF6". The identification of targets of AA263 and its analogs is clear. However, it is a presumption that the overlap in PDIs as targets of both AA263 and AA147 means that AA263 works through the PDIs. While a likely mechanism, this conclusion would be bolstered by establishing that knockdown of the PDIs lessens drug impact with respect to ATF6 activation.

We thank the reviewer for this comment. We previously showed that genetic depletion of different PDIs modestly impacts ATF6 activation afforded by ATF6 activating compound such as AA147 (see Paxman et al (2018) ELIFE). However, as discussed in this manuscript, the

ability for AA147 and AA263 to activate ATF6 signaling is mediated through polypharmacologic targeting of multiple different PDIs involved in regulating the redox state of ATF6. Thus, individual knockdowns are predicted to only minimally impact the ability for AA263 and its analogs to activate ATF6 signaling.

To address this comment, we have tempered our language regarding the mechanism of AA263-dependent ATF6 activation through PDI targeting described herein to better reflect the fact that we have not explicitly proven that PDI targeting is responsible for this activity, as highlighted below:

“Page 7, Line 158: “Intriguingly, 12 proteins were shared between these two conditions, including 7 different ER-localized PDIs (Fig. 1H). This includes PDIs previously shown to regulate ATF6 activation including TXNDC12/ERP18.[45,46] These results are similar to those observed when comparing proteins modified by the selective ATF6 activating compound AA147^{yne} and AA132^{yne}. [38] Further, we found that the extent of labeling for PDIs including PDIA1, PDIA4, PDIA6, and TMX1, but not TXNDC12, showed greater modification by AA132^{yne}, as compared to AA263^{yne} (Fig. 1I). Similar results were observed for AA147^{yne}. [38] This suggests that, like AA147, the selective activation of ATF6 afforded by AA263 is likely attributed to the modifications of a subset of multiple different ER-localized PDIs by this compound.”

Alternatively, it has previously been suggested that the cell-type dependent activity of AA263 may be traced to the presence of cell-type specific P450s that allow for the metabolic activation of AA263 or cell-type specific PDIs (Plate et al 2016; Paxman et al 2018). If the PDI target profile is distinct in different cell types, and these target difference correlates with ATF6-induced activity by AA263, that would also bolster the authors' conclusion.

As highlighted by the reviewer, different ER oxidases (e.g., P450s) could differentially influence activation of compounds such as AA263 to promote PDI modification and subsequent ATF6 activation. The specific ER oxidases responsible for AA263 activation are currently unknown; however, we anticipate that multiple different enzymes can promote this activity making it difficult to discern the specific contributions of any one oxidase. We have made this point clearer in the revised submission, as below:

Page 7, Line 169: “This specificity for ER proteins instead suggests the localized generation of AA263 quinone methides at the ER membrane, likely through metabolic activation by different ER localized oxidases, which has been previously been shown to contribute to the selective modification of ER proteins afforded by other compounds such as AA147 [49]”

Reviewer #2 (Public review):

Modulating the UPR by pharmacological targeting of its sensors (or regulators) provides mostly uncharted opportunities in diseases associated with protein misfolding in the secretory pathway. Spearheaded by the Kelly and Wiseman labs, ATF6 modulators were developed in previous years that act on ER PDIs as regulators of ATF6. However, hurdles in their medicinal chemistry have hampered further development. In this study, the authors provide evidence that the small molecule AA263 also targets and covalently modifies ER PDIs, with the effect of activating ATF6. Importantly, AA263 turned out to be amenable to chemical optimization while maintaining its desired activity. Building on this, the authors show that AA263 derivatives can improve the aggregation, trafficking, and function of two disease-associated mutants of secretory pathway proteins. Together, this study provides compelling evidence for AA263 (and its derivatives) being interesting modulators of ER proteostasis. Mechanistic details of its mode of action will need more attention in future studies that can now build on this.

We thank the reviewer for their positive comments on our manuscript. We address the reviewer's specific queries on our work, as outlined below.

In detail, the authors provide strong evidence that AA263 covalently binds to ER PDIs, which will inhibit the protein disulfide isomerase activity. ER PDIs regulate ATF6, and thus their finding provides a mechanistic interpretation of AA263 activating the UPR. It should be noted, however, that AA263 shows broad protein labeling (Figure 1G), which may suggest additional targets, beyond the ones defined as MS hits in this study.

This is true. We do show broad proteome-wide labeling with AA263^{yne}, which are largely reflected in the hits identified by MS beyond PDI family members. It is possible that other observed engaged targets, in addition to PDIs, may contribute to the activation of ATF6 signaling. Regardless, our MS analysis clearly shows that the compounds modified by AA263 are enriched for PDIs, further supporting our model whereby AA263-dependent PDI modification is likely responsible for ATF6 activation.

Also, a further direct analysis of the IRE1 and PERK pathways (activated or not by AA263) would have been a benefit, as e.g., PDIA1, a target of AA263, directly regulates IRE1 (Yu et al., EMBOJ, 2020), and other PDIs also act on PERK and IRE1. The authors interpret modest activation of IRE1/PERK target genes (Figure 2C) as an effect on target gene overlap, indeed the most likely explanation based on their selective analyses on IRE1 (ERdj4) and PERK (CHOP) downstream genes, but direct activation due to the targeting of their PDI regulators is also a possible explanation.

While we do observe mild increases in IRE1/XBP1s target genes, we do not observe significant increases in PERK/ISR target genes in cells treated with optimized AA263 analogs (see Fig. 2C). We previously showed that genetic ATF6 activation leads to a modest increase in IRE1/XBP1s target genes, reflecting the overlap in target genes of the IRE1/XBP1s and ATF6 pathways (see Shoulders et al (2013) Cell Reports). However, with our data, we cannot explicitly rule out the possibility that the mild increase in IRE1/XBP1s target genes reflects direct IRE1/XBP1s activation, as suggested by the reviewer. To address this, we have adapted the text to highlight this point, now specifically referring to preferential ATF6 activation afforded by these compounds, as below:

Page 5, Line 100: "In addition to finding AA147, our original high-throughput screen also identified the phenylhydrazone compound AA263 as a compound that preferentially activates the ATF6 arm of the UPR [26]"

Further key findings of this paper are the observed improvement of AAT behavior and GABAA trafficking and function. Further strength to the mechanistic conclusion that ATF6 activation causes this could be obtained by using ATF6 inhibitors/knockouts in the presence of AA263 (as the target PDIs may directly modulate the behavior of AAT and/or GABAA).

AA263 and related compounds could influence ER proteostasis of destabilized proteins through multiple mechanisms including ATF6 activation or direct modification of a subset of PDIs. We previously showed that AA263-dependent enhancement of A1AT-Z secretion and activity can be largely attributed to ATF6 activation (see Sun et al (2023) Cell Chem Biol). In the revised submission, we now show that increased levels of g2(R177G) afforded by treatment with AA263^{yne} are partially blocked by co-treatment with the ATF6 inhibitor Ceapin-A7 (CP7), highlighting the contributions of ATF6 activation for this phenotype (Fig. S5B,C). Intriguingly, this result also demonstrates the benefit for targeting ER proteostasis using compounds such as our optimized AA263 analogs, as this approach allows us to enhance ER proteostasis of destabilized proteins through multiple mechanisms. We further expand on this specific point in the revised manuscript as below:

Page 14, Line 375: “AA263 and its related analogs can influence ER proteostasis in these models through different mechanisms including ATF6-dependent remodeling of ER proteostasis and direct alterations to the activity of specific PDIs.(*) Consistent with this, we show that pharmacologic inhibition of ATF6 only partially blocks increases of g2(R177G) afforded by treatment with AA263^{yne}, highlighting the benefit for targeting multiple aspects of ER proteostasis to enhance ER proteostasis of this diseaserelevant GABA_A variant. While additional studies are required to further deconvolute the relative contributions of these two mechanisms on the protection afforded by our optimized compounds, our results demonstrate the potential for these compounds to enhance ER proteostasis in the context of different protein misfolding diseases.”

Along the same line, it also warrants further investigation why the different compounds, even if all were used at concentrations above their EC50, had different rescuing capacities on the clients.

This is an interesting question that we are continuing to study. While in general, we observe fairly good correlation between ATF6 activation and correction of diseases of ER proteostasis linked to proteins such as A1AT-Z or GABA_A receptors, as the reviewer points out, we do find some compounds are more efficient at correcting proteostasis than others activate ATF6 to similar levels. We attribute this to differences in either labeling efficiency of PDIs or differential regulation of various ER proteostasis factors, although that remains to be further defined. As we continue working with these (and other) compounds, we will focus on defining a more molecular basis for these findings.

Together, the study now provides a strong basis for such in-depth mechanistic analyses.

We agree and we are continuing to pursue the mechanistic basis of ER proteostasis remodeling afforded by these and related compounds.

Reviewer #3 (Public review):

Summary:

This study aims to develop and characterize phenylhydrazone-based small molecules that selectively activate the ATF6 arm of the unfolded protein response by covalently modifying a subset of ER-resident PDIs. The authors identify AA263 as a lead scaffold and optimize its structure to generate analogs with improved potency and ATF6 selectivity, notably AA263-20. These compounds are shown to restore proteostasis and functional expression of disease-associated misfolded proteins in cellular models involving both secretory (AAT-Z) and membrane (GABA_A receptor) proteins. The findings provide valuable chemical tools for modulating ER proteostasis and may serve as promising leads for therapeutic development targeting protein misfolding diseases.

Strengths:

- (1) The study presents a well-defined chemical biology framework integrating proteomics, transcriptomics, and disease-relevant functional assays.*
- (2) Identification and optimization of a new electrophilic scaffold (AA263) that selectively activates ATF6 represents a valuable advance in UPR-targeted pharmacology.*
- (3) SAR studies are comprehensive and logically drive the development of more potent and selective analogs such as AA263-20.*
- (4) Functional rescue is demonstrated in two mechanistically distinct disease models of protein misfolding-one involving a secretory protein and the other a membrane protein-underscoring the translational relevance of the approach.*

We thank the reviewer for their positive comments related to our work. We address specific weaknesses highlighted by the reviewer, as outlined below.

Weaknesses:

(1) ATF6 activation is primarily inferred from reporter assays and transcriptional profiling; however, direct evidence of ATF6 cleavage is lacking.

While ATF6 trafficking and processing can be visualized in cell culture models following severe ER insults (e.g., Tg, Tm), we showed previously that the more modest activation afforded by pharmacologic activators such as AA147 and AA263 cannot be easily visualized by monitoring ATF6 processing (see Plate et al (2016) ELIFE). As we have shown in numerous other manuscripts, we have established a transcriptional profiling approach that accurately defines ATF6 activation. We use that approach to confirm preferential ATF6 activation in this manuscript. We feel that this is sufficient for confirming ATF6 activation. However, we also now include data showing that co-treatment with ATF6 inhibitors (e.g., CP7) blocks increased expression of ATF6 target genes induced by our prioritized compound AA263^{yne} (Fig. S1B). This further supports our assertion that this compound activates ATF6 signaling.

(2) While the mechanism involving PDI modification and ATF6 activation is plausible, it remains incompletely characterized.

We thank the reviewer for this comment. We previously showed that genetic depletion of different PDIs modestly impacts ATF6 activation afforded by ATF6 activating compound such as AA147. However, as discussed in this manuscript, the ability for AA147 and AA263 to activate ATF6 signaling is mediated through polypharmacologic targeting of multiple different PDIs involved in regulating ATF6 redox. Thus, individual knockdowns are predicted to only minimally impact the ability for AA263 and its analogs to activate ATF6 signaling.

To address this comment, we have tempered our language regarding the mechanism of AA263-dependent ATF6 activation through PDI targeting described herein to better reflect the fact that we have not explicitly proven that PDI targeting is responsible for this activity, as highlighted below:

Page 7, Line 158: “Intriguingly, 12 proteins were shared between these two conditions, including 7 different ER-localized PDIs (Fig. 1H). This includes PDIs previously shown to regulate ATF6 activation including TXNDC12/ERP18.[45,46] These results are similar to those observed when comparing proteins modified by the selective ATF6 activating compound AA147^{yne} and AA132^{yne}. [38] Further, we found that the extent of labeling for PDIs including PDIA1, PDIA4, PDIA6, and TMX1, but not TXNDC12, showed greater modification by AA132^{yne}, as compared to AA263^{yne} (Fig. 1I). Similar results were observed for AA147^{yne}[38] This suggests that, like AA147, the selective activation of ATF6 afforded by AA263 is likely attributed to the modifications of a subset of multiple different ER-localized PDIs by this compound.”

(3) No in vivo data are provided, leaving the pharmacological feasibility and bioavailability of these compounds in physiological systems unaddressed.

We are continuing to test the in vivo activity of these compounds in work outside the scope of this initial study.

Reviewer #1 (Recommendations for the authors):

(1) First page of the discussion, last sentence. "We previously showed the relatively labeling of PDI modification directly impacts..." should be reworded.

Thank you. We have corrected this in the revised manuscript.

(2) What is the rationale for measuring ERSE-Fluc activity at 18 h but RNAseq at 6 h? What is known about the timing of action for AA263?

Compound-dependent activation of luciferase reporters requires the translation and accumulation of the luciferase protein for sufficient signal, while qPCR does not. We normally use longer incubations for reporter assays to ensure that we have sufficient quantity of reporter protein to accurately monitor activation. We have found that AA263 can rapidly increase ATF6 activity, with gene expression increases being observed after only a few hours of treatment. This is consistent with the proposed mechanism of ATF6 activation discussed herein involving metabolic activation and subsequent PDI modification.

(3) Figure 1 panel E and Figure S2 panel B. Are these the same data for AA263 and AA263^{yne}, with the AA2635 added to the plot for Figure S2? If so, it would be nice to note that panel B represents data from 3 of the replicates that are shown in Figure 1 (n=6).

Yes. The AA263 and AA263^{yne} data shown in Fig. 1E and Fig. S2B are the same data, as these experiments were performed at the same time. We apologize for this oversight, which has now been corrected in the revised version. Note that there were n=3 replicates for the dose response shown in Fig. 1E, which we corrected in the figure legend as below:

Fig. S2B Figure Legend: “B. Activation of the ERSE-FLuc ATF6 reporter in HEK293T cells treated for 18 h with the indicated concentration of AA263, AA263^{yne}, or AA263-5. Error bars show SEM for n= 3 replicates. The data for AA263 and AA263^{yne} is the same as that shown in Fig. 1E and are shown for comparison.”

(4) Figure S3. The legend notes 5 μ M AA263-yne and 20 μ M analog, whereas the figure itself outlines the same ratio but different concentrations: 10 μ M and 40 μ M.

We apologize for this mistake in the legend, which has been corrected. The information in the figure is correct.

Reviewer #2 (Recommendations for the authors):

(1) The activation mechanism of ATF6 is still debated (really trafficking as a monomer?); the authors may want to word more carefully here.

We agree. We have corrected this in the revised manuscript to indicate that increased populations of reduced ATF6 traffic for proteolytic processing.

(2) In Figure 1B, below the figure, mM is written for BME, but micromolar is meant.

Thank you. This has been corrected in the revised manuscript.

(3) The authors may want to make clearer, why BME does not completely inhibit AA263 and does not cause ER stress itself under the conditions tested.

The addition of BME in our experiments is designed to shift the redox potential of the cell to increase intracellular thiol reagents, such as glutathione, that can quench ‘activated’ AA263 and its analogs. However, BME is actively being oxidized upon addition and the intracellular redox environment can rapidly equilibrate following BME addition. Thus, we do not expect that AA263 or other metabolically activated compounds will be fully quenched using this approach, as is observed. This is consistent with other experiments where we show that the use of these types of reducing agents do not fully suppress the activity of reactive molecules, instead shifting their dosedependent activation of specific pathways.

(4) The data in Figure 4C seems to disagree with the other data on the tested compounds; this should be clarified.

It is unclear to what the reviewer is referring. The data in 4C shows that treatment with our optimized AA263 analogs improved elastase inhibition afforded by secreted A1AT, as would be predicted.

(5) PDIs that have been shown to regulate ATF6 should be discussed in more detail in the light of the presented data/interactome (e.g., ERp18).

Thank you for the suggestion. We now explicitly note that AA263^{yne} covalently modifies TXNDC12/ERP18 in our proteomic dataset. However, we also note that there is no difference in labeling of this specific PDI between AA263^{yne} and AA132^{yne}. This may indicate that the targeting of this protein is responsible for the larger levels of ATF6 activation afforded by both these compounds relative to AA147, with the activation of other UPR pathways afforded by AA132 resulting from increased labeling of other PDIs. We are now exploring this possibility in work outside the scope of this current manuscript.

Page 7 Line 158: “Intriguingly, 12 proteins were shared between these two conditions, including 7 different ER-localized PDIs (Fig. 1H). This includes PDIs previously shown to regulate ATF6 activation including TXNDC12/ERP18.[45,46] These results are similar to those observed when comparing proteins modified by the selective ATF6 activating compound AA147^{yne} and AA132^{yne}. [38] Further, we found that the extent of labeling for PDIs including PDIA1, PDIA4, PDIA6, and TMX1, but not TXNDC12, showed greater modification by AA132^{yne}, as compared to AA263^{yne} (Fig. 1I). Similar results were observed for AA147^{yne} [38] This suggests that, like AA147, the selective activation of ATF6 afforded by AA263 is likely attributed to the modifications of a subset of multiple different ER-localized PDIs by this compound.”

Reviewer #3 (Recommendations for the authors):

(1) Please consider adding detection of ATF6 cleavage by Western blot as direct evidence of AA263-induced ATF6 activation, to substantiate the central mechanistic claim.

While ATF6 trafficking and processing can be visualized in cell culture models following severe ER insults (e.g., Tg, Tm), we showed previously that the more modest activation afforded by pharmacologic activators such as AA147 and AA263 cannot be easily visualized through monitoring ATF6 proteolytic processing by western blotting (see Plate et al (2016) ELIFE). As we have shown in numerous other manuscripts, we have established a transcriptional profiling approach that accurately defines ATF6 activation. We use that approach to confirm preferential ATF6 activation in this manuscript. We feel that this is sufficient for confirming ATF6 activation. However, we also now include qPCR data showing that co-treatment with ATF6 inhibitors (e.g., CP7) blocks increased expression of ATF6 target genes induced by our prioritized compounds.

(2) To strengthen causal inference, loss-of-function experiments such as PDI knockdown, cysteine mutant inactivation, or reconstitution studies may be informative.

We thank the reviewer for this comment. We previously showed that genetic depletion of different PDIs modestly impacts ATF6 activation afforded by ATF6 activating compound such as AA147. However, as discussed in this manuscript, the ability for AA147 and AA263 to activate ATF6 signaling is mediated through polypharmacologic targeting of multiple different PDIs involved in regulating ATF6 redox state rather than a single PDI family member. Thus, individual knockdowns are predicted to only minimally impact the ability for AA263 and its analogs to activate ATF6 signaling.

To address this comment, we have tempered our language regarding the mechanism of AA263-dependent ATF6 activation through PDI targeting described herein to better reflect the fact that we have not explicitly proven that PDI targeting is responsible for this activity.

(3) Since β -mercaptoethanol inhibits ATF6 activation, it would be helpful to examine whether DTT also suppresses the activity of AA263 or its analogs, to clarify the redox sensitivity of the mechanism.

The use of reducing agents stronger than BME, such as DTT, globally activates the UPR, including the ATF6 arm of the UPR. Thus, we are unable to perform the requested experiments. We specifically use BME because it is a sufficiently mild reducing agent that can quench reactive metabolites (e.g., activated AA263 analogs) through alterations in cellular glutathione levels without globally activating the UPR.

(4) Given the electrophilic nature of AA263, which may allow it to react with endogenous thiols (e.g., glutathione or cysteine), a brief discussion or experimental validation of this potential liability would enhance the interpretation of in vivo applicability.

Metabolically activated AA263, like AA147, can be quenched by endogenous thiols such as glutathione. However, treatment with our metabolically activatable electrophiles AA147 and AA263, either in vitro or in vivo, does not seem to induce activation of the NRF2-regulated oxidative stress response (OSR) in the cell lines used in this manuscript (e.g., Fig. S2C). This suggests that treatment with these compounds does not globally disrupt the intracellular redox state, at least in the tested cell lines. While AA147 has been shown to activate NRF2 in specific neuronal cell lines and in primary neurons, AA147 does not activate NRF2 signaling in other nonneuronal cell lines or other tissues (see Rosarda et al (2021) ACS Chem Bio). We are currently testing the potential for AA263 to similarly activate adaptive NRF2 signaling in neuronal cells. Regardless, AA147, which functions through a similar mechanism to that proposed for AA263, has been shown to be beneficial in multiple models of disease both in vitro and in vivo. This indicates that this mechanism of action is suitable for continued translational development to mitigate pathologic ER proteostasis disruption observed in diverse types of human disease.

(5) Evaluation of in vivo activity, such as BiP induction in the liver following intraperitoneal administration of AA263-20 or related analogs, could substantially increase the translational impact of the work.

We are continuing to probe the activity of our optimized AA263 analogs in vivo in work outside the scope of this current manuscript. We thank the reviewer for this suggestion.

(6) The degree of BiP induction may also be contextualized by comparison with known ER stress inducers such as thapsigargin or tunicamycin, ideally by providing relative dose-equivalent responses.

We are not sure to what the reviewer is referring. We show comparative activation of ATF6 in cells treated with the ER stressor Tg and our compounds by both reporter assay (e.g., Fig. 2B) and qPCR of the ATF6 target gene BiP (HSPA5) (Fig. S2A). We feel that this provides context for the more physiologic levels of ATF6 activation afforded by these compounds.

<https://doi.org/10.7554/eLife.107000.2.sa0>